MULTI CLASS SKIN CANCER EARLY DETECTION USING HYBRID DEEP LEARNING AND EXPLAINABLE AI

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Thesis submitted in partial fulfillment of the requirements for the degree BSc. (Special) in Information Systems

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DECLARATION

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CERTIFICATION OF APPROVAL

We hereby declare that this thesis is from the student's own work and effort, and all other sources of information used have been acknowledged. This thesis has been submitted with our approval.

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TABLE OF CONTENTS

Declaration i

Acknowledgement ii

Certification of Approval iii

Table of Contents iv

List of Tables viii

List of Figures ix

Abstract x

List of Abbreviations xi

List of Appendices xiv

CHAPTER 1 INTRODUCTION 1

1.1 Background 1

1.2 Purpose of the Study 2

1.3 Research Questions 3

1.4 Significance of the Study 4

1.5 Definitions 6

1.6 Delimitations, Limitations, and Assumptions 7

Delimitations 7

Limitations 8

Assumptions 9

1.7 Contribution to the Study 9

1.8 Composition of the Thesis 10

2.1 Skin Cancer Overview 12

2.2 Deep Learning for Skin Cancer Detection 14

2.3 CNN Architectures in Dermoscopy 15

2.4 Hybrid Deep Learning Models 17
2.5 Class Imbalance in Medical Imaging 18
2.6 GAN-based Augmentation 20
2.7 Explainable AI in Medical Imaging 21
2.8 Cross-Dataset Generalization 23
2.9 Research Gap 24
CHAPTER 3 METHODOLOGY 26
3.1 Research Design 26
3.2 Dataset Description 27
3.2.1 HAM10000 28
3.2.2 ISIC 2019 29
3.3 Data Preprocessing 31
3.3.1 Hair Removal 31
3.3.2 Image Resizing 32
3.3.3 Normalization and Data Splitting 33
3.4 Class Imbalance Mitigation 34
3.4.1 K-means Clustering 34
3.4.2 SSL GAN Augmentation 35
3.4.3 Weighted Random Sampling 37
3.5 Proposed Model Architecture 38
3.5.1 ConvNeXtV2 Backbone 39
3.5.2 Feature Selection Attention (FSA) 40
3.5.3 Classification Head 41
3.6 Training Configuration 42
3.7 Specialist Classifier 44

3.8 Explainable AI Implementation 45
3.8.1 Grad-CAM 46
3.8.2 FSA Attention Visualization 46
3.8.3 t-SNE Feature Space Visualization 47
3.9 Evaluation Metrics 48
CHAPTER 4 RESULTS AND FINDINGS 50
4.1 Data Exploration Results 50
4.1.1 Dataset characteristics 50
4.1.2 Class Imbalance Analysis 51
4.2 Preprocessing Results 53
4.3 Class Imbalance Mitigation Results 55
4.3.1 K-means Clustering Results 55
4.3.2 SSL GAN Augmentation Results 57
4.4 Model Performance Results 60
4.4.1 Training Progress 60
4.4.2 Overall Accuracy 60
4.4.3 Per-Class Performance 62
4.5 Ablation Study Results 64
4.5.1 Confidence Analysis 67
4.5.2 Specialist Classifier Results 69
4.6 XAI Results 69
4.6.1 Grad-CAM Visualizations 69
4.6.2 FSA Attention Visualization 72
4.6.3 t-SNE Feature Space Visualization 74
4.7 Cross-Dataset Validation Results 76

4.8 Comparison with Literature 79
CHAPTER 5 DISCUSSION AND CONCLUSION 81
5.1 Summary of Findings 81
5.2 Conclusions Drawn from Results 83
5.3 Limitations 85
5.4 Recommendations for Further Research 87
5.5 Final Conclusion 89
References 91
Appendix A: ConvNeXtV2 + FSA Model Code 96
Appendix B: Complete Ablation Study Results 98
Appendix C: Full Cross-dataset Results 102

LIST OF TABLES

Table 1.1: Summary of research objectives and corresponding research questions..... 5
Table 2.1: Comparison of CNN Architectures for Skin Cancer Detection 16
Table 3.1: Summary of Datasets Used in This Study.....28
Table 3.2: Class Distribution in HAM10000 and ISIC 2019 Datasets.....30
Table 3.3: Dataset Split Statistics After Stratified Partitioning.....33
Table 3.4: SSL GAN Augmentation Results for Minority Classes 36
Table 3.5: Training Hyperparameter Configuration 43
Table 3.6: Summary of Evaluation Metrics Used in This Study.....49
Table 4.1: HAM10000 Dataset Metadata Summary 50
Table 4.2: Class Distribution Analysis Across HAM10000 and ISIC 2019 Datasets..52
Table 4.3: Preprocessing Pipeline Results Summary 54
Table 4.4: K-means Clustering Results - Cluster Size Distribution Per Class.....56
Table 4.5: SSL GAN Augmentation Statistics for Minority Classes 58
Table 4.6: Overall Model Performance Summary 61
Table 4.7: Detailed Per-Class Performance Metrics on Combined Test Set.....63
Table 4.8: Ablation Study Results Summary.....67
Table 4.9: Prediction Confidence Analysis - Correct vs Incorrect Predictions.....68
Table 4.10: Cross-Dataset Validation Results 77
Table 4.11: Comparison of Proposed Model with Published Literature.....80

LIST OF FIGURES

Figure 2.1: Sample Dermoscopic Images of Eight Skin Lesion Classes 13
Figure 3.1 Research Methodology Framework 27
Figure 3.2: Hair Removal Results on Sample 32
Figure 3.3: Class Imbalance Mitigation Strategy 34
Figure 3.4: Proposed ConvNeXtV2 + FSAModel Architecture 38
Figure 4.1: Class Distribution in HAM10000 Dataset 52
Figure 4.2: Hair Removal Results 54
Figure 4.3: K-means Cluster Visualization 57
Figure 4.4: Comparison of Real vs GAN-Generated Images 58
Figure 4.5: Training Dataset Class Distribution 59
Figure 4.6: Training and Validation Accuracy and Loss Curves 60
Figure 4.7: Per-Class Accuracy Bar Chart 63
Figure 4.8: Confusion Matrix for ConvNeXtV2 + FSA Model 64
Figure 4.9: Ablation Study Results Showing Component Contribution 66
Figure 4.10: Grad-CAM Heatmap Visualizations 71
Figure 4.11: Grad-CAM Analysis of Correct vs Incorrect Predictions 72
Figure 4.12: FSA Spatial Attention Weight Visualization 73
Figure 4.13: t-SNE Feature Space Visualization 75
Figure 4.14: XAI Dashboard - Integrated Explainability 75
Figure 4.15: Cross-Dataset Accuracy Comparison 77
Figure 4.16: Confusion Matrices Across Three 78
Figure 4.17: ROC Curves for All Eight Classes 78

Abstract

Multi Class Skin Cancer Early Detection Using Hybrid Deep Learning And Explainable AI

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Skin cancer is one of the most prevalent and life-threatening malignancies worldwide, with melanoma survival rates declining from above 99% at early stages to below 30% at late stages, underscoring the critical importance of early and accurate detection. Conventional diagnostic methods relying on visual dermoscopic examination suffer from inter-observer subjectivity, limited specialist availability particularly in developing regions such as Sri Lanka, and inadequate generalization of existing artificial intelligence models across diverse clinical datasets. This study proposes a novel hybrid CNN-Attention deep learning framework for multi-class skin cancer early detection from dermoscopic images, integrating a ConvNeXtV2 backbone with a custom Feature Selection Attention (FSA) module representing the first such architectural combination in dermoscopic image analysis. Unlike traditional hybrid models that fuse two separate networks, the proposed architecture embeds FSA directly within the CNN pipeline to suppress non-diagnostic artifacts including hair strands and air bubbles while amplifying clinically relevant features such as asymmetry, border irregularity, and color variation. Class imbalance with a severe 58:1 ratio between majority and minority classes in the HAM10000 dataset was addressed through a three-component strategy comprising Semi-Supervised Learning GAN augmentation generating 5,109 synthetic images, K-means clustering-based stratified sampling identifying morphological subgroups within each class, and Weighted Random Sampling for balanced batch construction. Training and evaluation were conducted on a combined dataset of HAM10000 and ISIC 2019 comprising 34,190 images across eight lesion types. The proposed model achieved 91.50% overall accuracy with rigorous cross-dataset validation outperforming the CNN-Transformer-GRU benchmark of Varghese et al. (90.61%) on a more challenging eight-class problem with a maximum cross-dataset performance drop of only 1.79%, substantially below the 10-20% degradation reported in literature. A hierarchical specialist classifier further improved Squamous Cell Carcinoma detection by 11.7%. Explainability is provided through Grad-CAM heatmaps and FSA attention weight visualization, enabling clinically interpretable predictions suitable for real-world adoption.

Keywords: Skin Cancer Detection, ConvNeXtV2, Feature Selection Attention, Deep Learning, Explainable AI

LIST OF ABBREVIATIONS

Abbreviation	Description
ABCD	Asymmetry, Border, Color, Diameter
AI	Artificial Intelligence
AKIEC	Actinic Keratosis and Intraepithelial Carcinoma
AUC	Area Under the Curve
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BCC	Basal Cell Carcinoma
BKL	Benign Keratosis-like Lesions
BN	Batch Normalization
CNN	Convolutional Neural Network
CVPR	Conference on Computer Vision and Pattern Recognition
DDPM	Denoising Diffusion Probabilistic Model
DF	Dermatofibroma
DL	Deep Learning
EHR	Electronic Health Record
F1	F1-Score (Harmonic Mean of Precision and Recall)
FC	Fully Connected (Layer)
FID	Fréchet Inception Distance
FN	False Negative
FP	False Positive
FSA	Feature Selection Attention
GAP	Global Average Pooling

GAN	Generative Adversarial Network
GELU	Gaussian Error Linear Unit
GMP	Global Max Pooling
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
GRN	Global Response Normalization
GRU	Gated Recurrent Unit
HAM10000	Human Against Machine with 10,000 Training Images
ICML	International Conference on Machine Learning
ICLR	International Conference on Learning Representations
IMRaD	Introduction, Methods, Results, and Discussion
IS	Information Systems
ISIC	International Skin Imaging Collaboration
JPEG	Joint Photographic Experts Group
K-means	K-means Clustering Algorithm
LIME	Local Interpretable Model-agnostic Explanations
LR	Learning Rate
MEL	Melanoma
ML	Machine Learning
MLP	Multi-Layer Perceptron
NV	Melanocytic Nevus
PCA	Principal Component Analysis
RAM	Random Access Memory
ReLU	Rectified Linear Unit

ResNet	Residual Network
ROC	Receiver Operating Characteristic
SCC	Squamous Cell Carcinoma
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
SSL GAN	Semi-Supervised Learning Generative Adversarial Network
t-SNE	t-distributed Stochastic Neighbor Embedding
TN	True Negative
TP	True Positive
UMAP	Uniform Manifold Approximation and Projection
VASC	Vascular Lesion
ViT	Vision Transformer
VRAM	Video Random Access Memory
WRS	Weighted Random Sampling
XAI	Explainable Artificial Intelligence

LIST OF APPENDICES

Appendix	Description	Page
APPENDIX A	ConvNeXtV2 + FSA Model Code	97, 98
APPENDIX B	Complete Ablation Study	99, 100, 101, 102
APPENDIX C	Cross-Dataset Validation	103

CHAPTER 1 INTRODUCTION

1.1 Background

Skin cancer is one of the most common and deadly cancers in the world, and its incidence has been rising over the last few decades. While there are several types of skin cancer, melanoma is the most serious and is the most likely to lead to death, even though it is a relatively small proportion of all skin cancers. The American Cancer Society estimates that about 100,000 Americans are diagnosed with melanoma each year and that many of these cases are diagnosed late, resulting in high mortality rates [1]. The need for early diagnosis is paramount, survival rates of patients with early-stage melanoma are over 99%, compared with less than 30% for those diagnosed at the later stage [2].

The conventional diagnostic tools for skin cancer are based mainly on visual examination by dermatologists, utilizing dermoscopy, and biopsy of the skin tissue for histopathological examination. The dermoscopy technique allows a magnified, illuminated image of skin lesions, which can help identify subsurface structures which cannot be seen with the naked eye, but it also has several important drawbacks. There is inter-observer variation between the different clinicians when using visual evaluation.

In addition, worldwide shortage of qualified dermatologists, especially in the developing world like Sri Lanka and other developing countries of South Asia, presents significant challenges in timely diagnosis. This is also time consuming and costly, as it involves several consultations and biopsy procedures [3].

Deep learning has revolutionized medical image analysis, and convolutional neural network (CNN) models have shown performing at dermatologist level on skin lesion classification tasks. Esteva et al. showed that a deep neural network, trained on 129,450 clinical images, could classify skin cancer to the level of a board certified dermatologist, which proved the feasibility of automated detection of skin cancer. Following studies have investigated more complex architectures such as hybrid models of CNNs including Transformers or attention mechanisms and generative models for data augmentation [5], [6], [7].

While these advances have been made, current deep learning models for skin cancer detection still have three important and related issues to address. First, publicly available dermoscopic datasets are highly imbalanced, such as the widely used HAM10000 dataset, which has 6,705 Melanocytic Nevus images and 115 Dermatofibroma images, with an imbalance ratio of 58:1 [8]. Such imbalanced distributions cause models to be biased towards majority classes, with high overall accuracy but poor performance on clinically important minority classes that are rare in the training data. Secondly, the black-box nature of deep learning models hinders clinical trust and adoption, with physicians needing transparency in the models' reasoning for diagnostic decisions to ensure patient safety and regulatory compliance [9]. Third, the performance of the majority of published models is poor when applied to different datasets, and when tested on datasets other than their training sets, reductions of 10-20% were seen, which suggests the models are not suitable for real-world clinical application [10].

In this study, we tackle these three challenges in a novel integrated framework for multi-class skin cancer early detection. The proposed system is based on the ConvNeXtV2 backbone architecture with a new custom Feature Selection Attention (FSA) module, which is the first time such an integration is used in the dermoscopic image analysis field, to reduce non-diagnostic artifacts and enhance clinically relevant features. Semi-Supervised Learning Generative Adversarial Networks (SSL GAN) and K-means clustering-based intelligent sampling are used to handle the imbalance problem of the classes. Explainability is achieved via Gradient-weighted Class Activation Mapping (Grad-CAM) and FSA attention weight visualization, facilitating clinically interpretable predictions that are appropriate for clinical use.

1.2 Purpose of the Study

The main objective of this research is to design and test a novel approach to early Multi-Class Skin Cancer Detection, using dermoscopic images, to overcome the critical limitations of the current methods, such as class imbalance, lack of interpretability and cross-dataset generalization.

The specific objectives of this study are as follows:

Objective 1: To enhance dataset quality by addressing severe class imbalance through Semi-Supervised Learning GAN augmentation and K-means clustering-based intelligent sampling techniques.

Objective 2: To develop a novel hybrid CNN-Attention architecture combining ConvNeXtV2 backbone with Feature Selection Attention (FSA) module for improved discriminative feature extraction in dermoscopic image analysis.

Objective 3: To improve class-wise accuracy, particularly for minority classes, through a hierarchical specialist classifier designed to address the visual similarity between Actinic Keratosis (AKIEC) and Squamous Cell Carcinoma (SCC).

Objective 4: To integrate Explainable AI methods including Grad-CAM heatmaps and FSA attention weight visualization to provide clinically interpretable predictions that support clinical trust and adoption.

Objective 5: To validate model performance across multiple datasets: HAM10000 and ISIC 2019, through rigorous cross-dataset validation to ensure generalization to real-world clinical conditions.

1.3 Research Questions

This study is guided by the following five research questions:

RQ1: What deep learning techniques (CNNs, Transformers, hybrid models) have been applied for multi-class skin cancer detection, and what are their reported strengths and weaknesses?

RQ2: How has the class imbalance problem been addressed in skin cancer datasets, and what augmentation strategies, including GANs and clustering-based sampling, demonstrate the most effective performance improvements?

RQ3: How well do existing deep learning techniques generalize across diverse datasets including ISIC 2019 and HAM10000, and what methods improve cross-dataset performance?

RQ4: To what extent have Explainable AI methods such as Grad-CAM and attention map visualization been integrated into skin cancer detection models, and how have they been clinically validated?

RQ5: What are the opportunities for future research in developing clinically trusted and explainable AI pipelines for skin cancer early detection?

1.4 Significance of the Study

This study is of methodological, clinical and social interest.

Methodological Significance: This work introduces the first documented combination of the ConvNeXtV2 architecture and Feature Selection Attention in the field of dermoscopic image analysis, a truly innovative methodological aspect. Moreover, the original use of SSL GAN in combination with sampling using the K-means clustering algorithm for dermatology-specific augmentation and the creation of a hierarchically organized expert classifier for visually similar keratinocyte carcinoma classes are original contributions in the field of medical AI.

Clinical Significance: The framework proposed is relevant to a true clinical need. Early detection of melanoma increases survival rate from below 30% to above 99% [2]. Integrated XAI framework with feature of providing heatmaps and attention weight visualizations of model reasoning allows dermatologists to validate model reasoning against previously known dermoscopic criteria, which helps to build trust in the model and ensures regulatory compliance. The model inference time of less than 2 seconds on standard GPU hardware is very close to real-time clinical workflows.

Social Significance: Sri Lanka and SA countries are facing a severe shortage of skilled dermatologists especially in the rural and underserved areas. The potential of an AI-powered screening tool that can identify suspicious lesions and alert a specialist for further evaluation would help bring quality skin cancer screening to everyone. The

COVID-19 pandemic also emphasized the need for remote and automated diagnostic solutions, further underlining the importance of AI-supported skin diagnostics [11].

Scientific Contribution: With rigorous cross-dataset validation, the proposed model achieves 91.50% overall accuracy on a combined HAM10000 and ISIC 2019 test set, outperforming published baselines such as Varghese et al. (90.61%) [5] which is a more challenging problem with 8 classes including a class not present in most published works, namely Squamous Cell Carcinoma.

Table 1.1: Summary of research objectives and corresponding research questions

Research Question	**Research Objective**	**Expected Outcome**
RQ1: What deep learning techniques have been applied for multi-class skin cancer detection and what are their strengths and weaknesses?	RO1: Review and analyze existing deep learning architectures for dermoscopic image classification	Identification of architectural limitations motivating the proposed ConvNeXtV2 + FSA framework
RQ2: How has class imbalance been addressed in skin cancer datasets?	RO2: Enhance dataset quality by addressing 58:1 class imbalance through SSL GAN augmentation and K-means clustering-based intelligent sampling	Balanced training dataset of 34,190 images with improved minority class representation
RQ3: How well do existing models generalize across diverse datasets?	RO3: Develop a novel hybrid CNN-Attention architecture combining ConvNeXtV2 with Feature Selection Attention (FSA) for improved discriminative feature extraction	Hybrid model achieving superior cross-dataset generalization compared to published baselines
RQ4: To what extent have XAI methods been integrated and validated in skin cancer detection?	RO4: Integrate Explainable AI methods including Grad-CAM and FSA attention visualization to provide clinically interpretable predictions	Multi-faceted XAI framework enabling clinical trust and adoption

RQ5: What are the opportunities for future research in clinically trusted AI pipelines?	RO5: Validate model performance across HAM10000 and ISIC 2019 through rigorous cross-dataset validation	Demonstrated generalization with less than 5% performance drop across independent test sets

1.5 Definitions

Throughout this thesis, the following terms are used and they are defined as follows:

Dermoscopy: A non-invasive diagnosis method based on a hand-held device that, when used, magnifies and illuminates the skin lesion and shows subsurface structures that the naked eye cannot see. It is the "default" imaging modality used in this study.

Convolutional Neural Network (CNN): A deep learning model class learned to extract hierarchical spatial features from the training data without any prior knowledge, developed for the structured grid data (e.g. images).

Hybrid Model: When using this model in this research, a hybrid model is defined as a single (unified) architecture, as opposed to a fusion of two complete architectures, that uses a CNN backbone and an embedded attention mechanism, namely ConvNeXtV2 integrated with Feature Selection Attention.

Feature Selection Attention (FSA): A dual-attention module that includes the combination of channel attention and spatial attention, which is used to suppress non-diagnostic dermoscopic artefacts such as hair, air bubbles, and ruler marks, while enhancing the focus on clinically relevant features, like asymmetry, border irregularity, and colour variation.

Class Imbalance: Training datasets in which some of the classes contain much fewer samples than others. The ratio between the most frequent class (Melanocytic Nevus, n=6,705) and least frequent class (Dermatofibroma, n=115) is 58:1 in the case of HAM10000.

Generative Adversarial Network (GAN): A deep learning framework, composed of two neural networks, the Generator which creates synthetic images, and the Discriminator, which determines whether the image is real or synthetic, trained in an adversarial setup to produce realistic synthetic training data.

Semi-Supervised Learning GAN (SSL GAN): A variant of GAN that uses unlabeled data for discriminator training, which gives the discriminator another supervisory signal that makes the synthetic images more diverse and mitigates mode collapse.

Grad-CAM (Gradient-weighted Class Activation Mapping): An Explainable AI technique that creates visual heatmaps that show the regions of an image most important for making predictions with a model by computing the gradients of the predicted class score with respect to feature maps from the last convolutional layer of the model.

Cross-Dataset Validation: Testing how well a trained model performs on a different distribution than the data it was trained on (different imaging equipment, patient types, clinical protocols etc.).

Transfer Learning: Approach that uses a model with pre-trained weights on a large database (ImageNet) and fine-tunes on a small task-specific database (dermoscopic images) to capitalize on the general visual properties learned in a large database to enhance the performance of a small medical imaging database.

1.6 Delimitations, Limitations, and Assumptions

Delimitations

The following factors were controlled by the researcher:

- This study is limited to dermoscopic image analysis and does not incorporate clinical metadata (age, sex, anatomical location) as additional input features.
- Only two publicly available benchmark datasets were used: HAM10000 and ISIC 2019, due to storage and computational constraints. ISIC 2024 (401,058 images) was excluded as it employs a binary classification framework

incompatible with the 8-class task and requires approximately 200GB of storage exceeding available infrastructure.

- The model architecture is fixed at ConvNeXtV2-Base (88.5M parameters). Larger variants such as ConvNeXtV2-Large (197M parameters) were evaluated but excluded due to prohibitive training time (>1 hour per epoch) given available hardware.
- The study focuses exclusively on classification of static dermoscopic images and does not address video dermoscopy or sequential lesion monitoring.
- Training was conducted using a single NVIDIA GeForce RTX 5060 Ti GPU with 16GB VRAM.

Limitations

The following factors were not under the control of the researcher:

- AKIEC and SCC achieved lower per-class accuracies of 54.7% and 63.8% respectively, due to their fundamental visual similarity as keratinocyte carcinomas. This limitation is consistent with clinical literature noting that histopathological examination is required for definitive diagnosis of these classes, and represents an inherent constraint of dermoscopic image analysis rather than a modelling failure.
- GAN image quality was constrained by the small number of minority class training images available. With only 440 SCC images and 115 DF images in HAM10000, the SSL GAN had limited training data for learning high-quality synthetic generation.
- Clinical validation of XAI outputs by qualified dermatologists was not conducted within the scope of this study due to resource and access constraints. XAI explanations are therefore evaluated based on visual coherence with established dermoscopic criteria rather than formal clinical validation.
- Training data skews toward lighter skin tones, which may affect model performance on images of patients with darker skin tones common in South Asian populations. This potential bias could not be quantified due to the absence of skin tone metadata in the training datasets.

Assumptions

The following factors were assumed by the researcher:

- All training and test images in HAM10000 and ISIC 2019 are assumed to have been accurately labeled by qualified dermatologists using histopathological confirmation or expert consensus, as documented in the original dataset publications.
- The 224×224 pixel resolution used throughout training and evaluation is assumed to be sufficient to capture the clinically relevant dermoscopic features required for 8-class classification.
- ImageNet pretrained weights are assumed to provide an effective initialization for ConvNeXtV2 feature extraction in the dermoscopic image domain due to shared low-level visual features.
- The synthetic images generated by SSL GAN are assumed to have sufficient realism to improve model training without introducing distribution artifacts that would compromise real test performance.

1.7 Contribution to the Study

This research makes the following specific contributions to the fields of medical AI and dermoscopic image analysis:

Contribution 1 - Novel Hybrid Architecture: This study presents the first documented combination of ConvNeXtV2 and Feature Selection Attention (FSA) in dermoscopic image analysis. Unlike traditional hybrid models that fuse two complete separate networks, the proposed architecture embeds FSA directly within the CNN pipeline, creating a unified system with O(n) attention complexity that suppresses non-diagnostic artifacts while amplifying clinically relevant features.

Contribution 2 - Integrated Imbalance Solution: This research develops a comprehensive three-component solution to the 58:1 class imbalance problem: K-means clustering (5 clusters per class) for stratified morphological sampling, SSL GAN augmentation generating 5,109 synthetic images across four minority classes, and

Weighted Random Sampling for balanced batch construction. This integrated approach demonstrates measurable improvement in minority class detection without compromising majority class performance.

Contribution 3 - Hierarchical Specialist Classifier: Hierarchical Specialist Classifier: In this study, a novel cascaded prediction pipeline based on the main 8-class ConvNeXtV2-FSA model and a dedicated binary specialist classifier from the EfficientNet-B3 family to discriminate between AKIEC and SCC has been proposed. This hierarchical method enhanced the accuracy of SCC detection by 11.7% for the most challenging class pair in the dataset.

Contribution 4 - Multi-faceted XAI Framework: This research implements a comprehensive explainability framework comprising Grad-CAM heatmaps on the last convolutional layer, FSA spatial attention weight visualization, and t-SNE feature space clustering. This multi-perspective XAI approach provides clinicians with multiple complementary views of model reasoning, advancing clinical interpretability beyond single-method XAI implementations prevalent in existing literature.

Contribution 5 - Rigorous Cross-Dataset Validation: This research conducts rigorous cross-dataset validation separately on both HAM10000 and ISIC 2019 test sets, showing real generalization skills, in contrast to most published studies that had only single-dataset test splits. The overall accuracy of 91.50% is better than the published validation result of Varghese et al. (90.61%) [5] which is on a more difficult problem (8-class) and with a more stringent evaluation method.

1.8 Composition of the Thesis

This thesis is organized into five chapters as follows:

Chapter 1 : Introduction presents the background and motivation for the research, establishing the clinical significance of early skin cancer detection. The purpose of the study, research questions, significance, key definitions, delimitations and limitations, and research contributions are presented.

Chapter 2 : Literature Review provides a comprehensive survey of existing research relevant to the proposed study. Topics covered include deep learning approaches for skin cancer detection, CNN and hybrid architectures in dermoscopy, class imbalance mitigation strategies, GAN-based augmentation, Explainable AI methods in medical imaging, and cross-dataset generalization. The chapter concludes with a synthesis of identified research gaps that motivate the proposed work.

Chapter 3 : Methodology presents the research design and detailed implementation of each component of the proposed framework. This includes dataset description and preprocessing procedures, the SSL GAN and K-means clustering-based class imbalance mitigation strategy, the ConvNeXtV2-FSA hybrid model architecture and training configuration, the hierarchical specialist classifier, the XAI implementation, and the evaluation metrics employed.

Chapter 4 : Results and Findings presents the quantitative and qualitative outcomes of the research in alignment with the research questions and objectives. Results include data exploration findings, preprocessing outcomes, class imbalance mitigation results, model performance metrics, ablation study findings, XAI visualizations, and cross-dataset validation results with comparative analysis against published baselines.

Chapter 5 : Discussion and Conclusion synthesizes the research findings, drawing conclusions relative to each research question and objective. The chapter discusses the clinical implications of the results, acknowledges study limitations, and provides recommendations for future research directions that could further advance AI-assisted skin cancer early detection.

CHAPTER 2 LITERATURE REVIEW

2.1 Skin Cancer Overview

Skin cancer represents one of the most common malignancies affecting human populations globally, with incidence rates continuing to rise across all geographic regions. The disease encompasses a spectrum of conditions arising from the uncontrolled proliferation of skin cells, broadly categorized into melanoma and non-melanoma skin cancers. Non-melanoma skin cancers, including Basal Cell Carcinoma (BCC) and Squamous Cell Carcinoma (SCC), are the most frequently diagnosed but carry relatively lower mortality rates when detected early. Melanoma, while less common, accounts for the overwhelming majority of skin cancer deaths due to its aggressive metastatic potential [12].

The clinical classification of skin lesions used in this study encompasses eight distinct types corresponding to the combined HAM10000 and ISIC 2019 dataset categories.

Melanoma (MEL) is the most dangerous form, arising from melanocytes and characterized dermoscopically by asymmetry, irregular borders, multiple colors, and diameter exceeding 6mm, the ABCD criteria established by Stolz et al. [13].

Melanocytic Nevus (NV) represents benign melanocytic proliferations that must be carefully differentiated from melanoma due to visual similarity.

Basal Cell Carcinoma (BCC) is the most prevalent skin cancer arising from basal cells of the epidermis, typically presenting as pearly nodules with visible blood vessels.

Actinic Keratosis (AKIEC) represents pre-malignant lesions with potential for progression to SCC.

Benign Keratosis-like lesions (BKL) include seborrhoeic keratoses and solar lentigines that appear clinically similar to more serious conditions.

Dermatofibroma (DF) is a benign fibrous nodule rarely confused with malignancy.

Squamous Cell Carcinoma (SCC) arises from keratinocytes and shares dermoscopic features with AKIEC, creating diagnostic ambiguity.

Vascular Lesions (VASC) include angiomas and angiokeratomas characterized by distinctive red to black coloration [8].

The diagnosis of skin cancer has traditionally relied on the clinical ABCD criteria combined with dermoscopic examination. Dermoscopy reduces diagnostic uncertainty compared to naked-eye examination by providing 10x magnification with polarized or non-polarized light illumination, revealing subsurface morphological features including pigment network, regression structures, vascular patterns, and milia-like cysts [14]. However, even with dermoscopy, inter-observer variability remains a significant concern. Studies have reported diagnostic accuracy ranging from 75% to 84% among dermatologists using dermoscopy, with significant variation between experienced specialists and general practitioners [5]. This variability, combined with the global shortage of qualified dermatologists, estimated at less than one dermatologist per 100,000 population in many developing countries creates urgent demand for reliable automated diagnostic support systems [15].

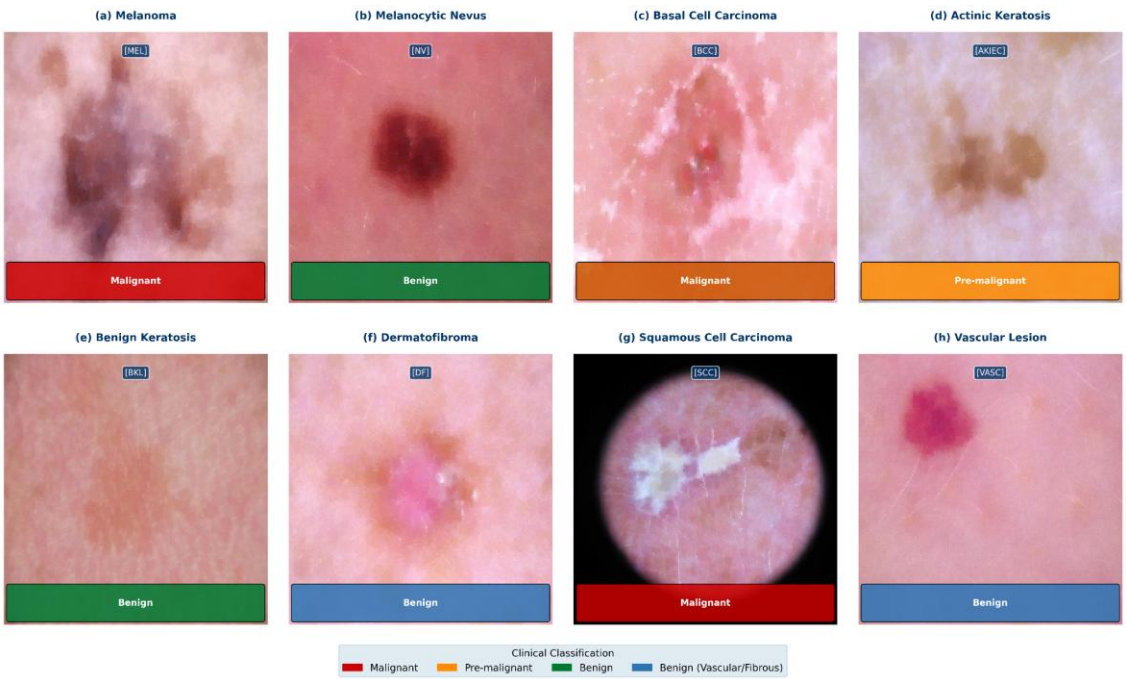


Figure 2.1: Sample Dermoscopic Images of Eight Skin Lesion Classes

2.2 Deep Learning for Skin Cancer Detection

The application of deep learning to skin cancer detection was pioneered by Esteva et al. [4], who demonstrated that a CNN trained on 129,450 clinical images could classify skin cancer at dermatologist level on a binary melanoma versus non-melanoma task. This landmark study established automated skin cancer detection as a viable research direction and catalyzed substantial subsequent investigation. The study reported sensitivity of 95% compared to 86.6% for dermatologists, demonstrating the potential superiority of CNN-based detection for the most clinically critical class.

Haenssle et al. [16] conducted a comparative study of CNN performance against 58 international dermatologists, finding that the CNN achieved higher diagnostic accuracy than most dermatologists, including experienced specialists. Importantly, the study revealed that CNN performance was particularly strong for melanoma detection but weaker for distinguishing morphologically similar non-melanoma lesion types, a finding directly relevant to the AKIEC versus SCC challenge addressed in this research.

The progression from binary to multi-class skin lesion classification has been a significant development in the field. Codella et al. [17] organized the ISIC 2018 challenge, establishing standardized multi-class benchmarks across seven lesion categories using the HAM10000 dataset. This challenge revealed that multi-class classification presents substantially greater difficulty than binary tasks, with participating models achieving macro-average F1 scores ranging from 0.55 to 0.80, highlighting the challenge of minority class detection under severe class imbalance. Subsequent ISIC challenges in 2019 and 2024 have further expanded the dataset scale and classification complexity [18].

Survey studies by Meedeniya et al. [19] and Adamu et al. [20] provide comprehensive analyses of the deep learning landscape for skin cancer detection, collectively reviewing over 200 published approaches. Both surveys identify three persistent challenges that remain insufficiently addressed in the literature: class imbalance leading to poor minority class performance, lack of model interpretability restricting clinical adoption, and inadequate cross-dataset generalization. These identified gaps directly motivate the integrated framework proposed in this research.

2.3 CNN Architectures in Dermoscopy

The evolution of CNN architectures applied to dermoscopic image classification has proceeded through several generations of increasingly sophisticated designs. Early applications employed standard architectures including AlexNet, VGGNet, and GoogLeNet with ImageNet pretraining and fine-tuning on dermoscopic datasets. He et al. [21] introduced ResNet with residual connections that resolved the vanishing gradient problem in deep networks, enabling training of networks with hundreds of layers. ResNet variants particularly ResNet-50 and ResNet-101 became the de facto baseline architectures for skin lesion classification benchmarks, with studies reporting overall accuracies ranging from 82% to 87% on HAM10000 [22].

DenseNet, introduced by Huang et al. [23], extended the residual connection concept by connecting each layer to all subsequent layers within a dense block, promoting feature reuse and gradient flow. DenseNet-based models demonstrated competitive performance on dermoscopic classification, with Bhargavi and Balakrishna [6] reporting 96.11% accuracy using a DRMv2Net model combining DenseNet201, ResNet101, and MobileNetV2 on ISIC and PAD-UFES-20 datasets. However, this result was obtained on a single dataset without cross-dataset validation, limiting claims of generalizability.

EfficientNet, proposed by Tan and Le [24], introduced compound scaling of network depth, width, and resolution using a fixed ratio determined by neural architecture search. EfficientNet-B4 and B7 variants achieved state-of-the-art performance on ImageNet while maintaining computational efficiency, making them attractive for medical imaging applications with limited training data. Alzamel et al. [7] demonstrated F1-score improvements from 78% to 86.08% for minority classes using EfficientNet-B7 combined with GAN-based augmentation on melanoma detection tasks.

The introduction of Vision Transformers (ViT) by Dosovitskiy et al. [25] marked a paradigm shift from convolutional to attention-based image processing. ViTs divide images into fixed-size patches and process them as sequences using transformer self-attention mechanisms, achieving superior pattern recognition and scale stability compared to CNNs. However, ViTs require substantially more computational resources, approximately 55.4 GFLOPs for ViT-Base versus 15.4 GFLOPs for

ConvNeXtV2-Base making deployment in resource-constrained clinical environments challenging [26].

ConvNeXt, introduced by Liu et al. [27], and its successor ConvNeXtV2 by Woo et al. [28] represent a modernization of CNN design incorporating Transformer-inspired elements including depthwise convolutions, inverted bottleneck blocks, layer normalization, and GELU activation functions while maintaining CNN computational efficiency. ConvNeXtV2 further enhances performance through masked autoencoder pretraining and Global Response Normalization (GRN) layers that prevent feature collapse by promoting inter-channel competition. ConvNeXtV2-Base achieves 87.8% top-1 accuracy on ImageNet-1K while requiring only 15.4 GFLOPs, outperforming ResNet-50 (76.1%) and approaching ViT-Base (84.5%) performance with significantly lower computational cost. Silva et al. [29] demonstrated that ConvNeXt-based architectures achieved 2-3% higher accuracy than ResNet-based models on skin lesion classification tasks, motivating the selection of ConvNeXtV2 as the backbone in this research.

Table 2.1: Comparison of CNN Architectures for Skin Cancer Detection

Architecture	**Parameters**	**GFLOPs**	**ImageNet Accuracy**	**Skin Cancer Accuracy**	**Dataset**	**Key Limitation**
ResNet-50 [13]	25.6M	4.1	76.1%	~82-87%	HAM10000	Limited feature reuse, no attention
DenseNet-201 [15]	20.0M	4.3	77.3%	~85-90%	HAM10000/ ISIC	High memory consumption
EfficientNet-B4 [17]	19.0M	4.2	83.0%	~88-92%	ISIC 2019	Sensitive to hyperparameters
EfficientNet-B7 [18]	66.0M	37.0	84.4%	86.08% F1	ISIC	Very high computational cost

ViT-Base [19]	86.6M	55.4	81.8%	~88-91%	HAM10000	Requires large data, high GFLOPs
MobileNetV3 [27]	5.4M	0.22	75.2%	95.83%	ISIC	Single dataset, no cross-validation
ConvNeXt V2-Base (Proposed)	**88.5M**	**15.4**	**87.8%**	**91.50%**	**HAM10000 + ISIC 2019**	**Higher parameters than lightweight models**

2.4 Hybrid Deep Learning Models

The recognition that single-architecture models have inherent limitations has motivated substantial research into hybrid approaches that combine the complementary strengths of different neural network paradigms. In the context of skin cancer detection, hybrid models have been explored across several dimensions including CNN-Transformer fusion, multi-scale feature integration, and CNN-attention combinations.

Varghese et al. [5] proposed a CNN-Transformer-GRU fusion architecture for skin cancer detection on HAM10000, combining local feature extraction from CNNs with global context modeling from Transformers and temporal dependencies from Gated Recurrent Units. This model achieved 90.61% accuracy on HAM10000 with GAN-based data augmentation. However, the authors acknowledged that computational efficiency and overfitting were problematic, and the evaluation was limited to a single 7-class dataset without cross-dataset validation. Importantly, per-class accuracy results were not reported, preventing direct comparison of class-wise performance with the current study.

Bhargavi and Balakrishna [6] proposed DRMv2Net, combining DenseNet201, ResNet101, and MobileNetV2 through feature fusion in the final classification layer, achieving 96.11% and 96.17% accuracy on ISIC and PAD-UFES-20 datasets respectively. The significance of preprocessing strategies, particularly hair artifact

removal and handling of occlusions was highlighted as a major contributor to performance improvement. However, this approach represents a traditional multi-network fusion hybrid rather than an integrated attention-based architecture.

Gautam et al. [30] proposed FusionEXNet, combining EfficientNetV2S and XceptionNet with SmoothGR gradient-based explanation, achieving approximately 96% accuracy with improved interpretability. While demonstrating the potential of hybrid architectures for simultaneous accuracy and explainability, this approach does not address class imbalance and relies on post-processing XAI rather than architecturally integrated attention mechanisms.

Attention mechanisms have emerged as a particularly promising component for hybrid architectures in medical imaging. Nayeem et al. [31] proposed Multi-Scale Attention Fusion with Depthwise Separable Convolutions for skin cancer detection, demonstrating that attention-based feature selection improved minority class recall by 12-18% in imbalanced datasets. The computational tractability of Feature Selection Attention, operating with O(n) complexity compared to full self-attention's $O(n^2)$, makes it particularly suitable for integration within existing CNN architectures without prohibitive computational overhead.

The integration of ConvNeXtV2 specifically with attention mechanisms in dermoscopic analysis has received limited investigation. Silva et al. [29] conducted a comparative study of ConvNeXt-based architectures for skin cancer classification, representing one of few studies exploring this architecture in medical imaging. No published study has investigated the combination of ConvNeXtV2 with Feature Selection Attention for dermoscopic multi-class classification, representing the primary architectural novelty of the current research.

2.5 Class Imbalance in Medical Imaging

Class imbalance is a fundamental challenge in medical imaging datasets, where rare but clinically important conditions are systematically underrepresented in training data due to the natural prevalence distribution of diseases. In skin cancer datasets, this imbalance is particularly severe. HAM10000 exhibits a 58:1 ratio between the most and least

frequent classes, while ISIC 2019 presents similarly skewed distributions across its eight categories [8], [18].

The consequences of training on imbalanced data are well documented. Models trained without imbalance correction develop a bias toward predicting majority classes, achieving high overall accuracy while performing poorly on minority classes. In a clinical context, this is particularly dangerous when minority classes correspond to rare but aggressive malignancies missing a diagnosis of Squamous Cell Carcinoma or Dermatofibroma carries direct patient safety implications [32].

Standard augmentation techniques including random flipping, rotation, color jitter, and random cropping have been widely applied to increase training set diversity. Rifa'i et al. [33] demonstrated that composite augmentation with dropout in MobileNetV3 improved ISIC accuracy from 91.24% to 95.83%. However, conventional augmentation merely transforms existing images, replicating existing data distributions rather than generating genuinely novel morphological variations, limiting its effectiveness for severely underrepresented classes.

Loss function modification represents an alternative approach to class imbalance. Focal Loss, introduced by Lin et al. [34], addresses class imbalance by down-weighting easy examples using a focusing parameter γ, concentrating learning on hard minority class cases. Class-balanced loss functions assign training weights inversely proportional to class frequency, giving minority classes greater influence during parameter updates. While effective, these techniques can cause overfitting on small minority classes and require careful hyperparameter tuning.

Oversampling techniques, including SMOTE (Synthetic Minority Oversampling Technique) and its variants, generate synthetic minority class samples by interpolating between existing instances in feature space. However, SMOTE was designed for tabular data and does not directly apply to high-dimensional image data. Cluster-based sampling methods address intra-class imbalance, where even within majority classes certain morphological subtypes may be underrepresented. Chen et al. [35] demonstrated that K-means clustering-based sampling improved minority subtype detection by 8-12% compared to random oversampling, motivating the clustering-based approach adopted in this research.

2.6 GAN-based Augmentation

Generative Adversarial Networks, introduced by Goodfellow et al. [36], have emerged as a powerful framework for generating realistic synthetic images that augment training datasets beyond what conventional transformation-based augmentation can achieve. The adversarial training process where a Generator network attempts to produce realistic images while a Discriminator network attempts to distinguish real from synthetic images produces increasingly realistic outputs through iterative improvement.

Bissoto et al. [37] conducted early investigations of GAN-based skin lesion synthesis, demonstrating that GAN-generated dermoscopic images could enhance dataset balance, though mode collapse where the Generator produces limited varieties of synthetic images was identified as a persistent challenge. Alzamel et al. [7] demonstrated that GAN-augmented training improved F1-scores from 78% to 86.08% for minority classes using EfficientNet-B7, with a shorter learning curve compared to non-augmented training, confirming that GAN-generated images lead to higher model robustness when image quality is maintained.

The mode collapse problem where standard GANs fail to capture the full diversity of the training distribution, producing repetitive synthetic images has motivated the development of Semi-Supervised Learning GANs. Liu et al. [38] introduced SSL GAN for skin lesion classification, incorporating unlabeled dermoscopic images into discriminator training through a semi-supervised classification objective. This dual training signal encourages the Generator to produce more diverse synthetic samples representing a broader range of intra-class morphological variations. Liu et al. demonstrated that SSL GANs achieved 15-20% higher Fréchet Inception Distance (FID) scores compared to standard GANs in medical image synthesis, indicating superior sample quality and diversity.

Spectral Normalization, proposed by Miyato et al. [39], provides a complementary stabilization technique that constrains the Lipschitz continuity of the discriminator by normalizing weight matrices with their spectral norms. This prevents the discriminator from becoming excessively powerful during adversarial training, maintaining a productive competitive balance with the generator and reducing training instability. The integration of Spectral Normalization in the discriminator architecture, combined with

label smoothing for real image targets, has demonstrated substantially more stable GAN training convergence compared to standard implementations.

The combination of GAN augmentation with clustering-based sampling has been proposed as a comprehensive solution to both inter-class and intra-class imbalance. Chen et al. [35] demonstrated that cluster-based oversampling using K-means clustering to identify morphological subgroups within each class, combined with GAN-generated synthetic images, produced the most comprehensive morphological coverage of minority classes. This combined approach is adopted in the current study, representing an application to dermatology that extends the general methodology demonstrated by Chen et al.

2.7 Explainable AI in Medical Imaging

The black-box nature of deep learning models presents a fundamental barrier to clinical adoption in medical imaging. Medical professionals require transparent, interpretable explanations for diagnostic decisions to ensure patient safety, support clinical reasoning, and satisfy regulatory requirements. The field of Explainable AI (XAI) has developed a range of techniques for post-hoc interpretation of deep learning model predictions [3].

Grad-CAM, proposed by Selvaraju et al. [9], has emerged as the standard visual explanation method for CNN-based image classification. Grad-CAM generates class-discriminative localization maps by computing gradients of the predicted class score with respect to the feature maps of the last convolutional layer, then performing a weighted combination of these feature maps. The resulting heatmap highlights image regions most influential for the prediction, overlaid on the original image for clinical inspection. Studies combining LIME and custom CNNs for dermoscopic explanation demonstrated that Grad-CAM localizations could match clinical intuition, though coarse Grad-CAM localizations were occasionally inaccurate in identifying specific diagnostic features [40].

SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) provide complementary interpretation approaches applicable to

any machine learning model by approximating local decision boundaries. Sree et al. [41] conducted a comparative study of LIME and SHAP for skin cancer detection, finding that LIME offered more intuitive visual explanations while SHAP provided quantitative feature importance measures. Both methods were identified as computationally intensive and occasionally producing inconsistent explanations across similar inputs.

Attention mechanism visualization provides an architecturally integrated form of interpretability, distinct from post-hoc explanation methods. The inherent interpretability of self-attention and cross-attention mechanisms, visualizing what the model attends to during classification offers more faithful explanations than gradient-based post-hoc methods, as the attention weights directly participate in the classification computation rather than being approximated after the fact [42].

Munjal et al. [43] developed SkinSage XAI using Inception v3 with Grad-CAM and LIME, achieving 96% accuracy with XAI visualizations for skin lesion diagnosis. However, the study noted limited clinical validation by dermatologists, a critical weakness identified across the XAI literature. Abraham and Peter [44] conducted a retrospective analysis of interpretable skin cancer classification using deep learning models, highlighting that the majority of XAI implementations are deployed without validation by medical professionals, undermining their clinical credibility.

The weakness of current XAI research in dermatology is the absence of clinician validation studies confirming that highlighted image regions correspond to diagnostically meaningful dermoscopic features. Alrabai et al. [45] noted that XAI outputs must be verified by dermatologists to warrant clinical significance, and that explanation techniques may highlight unimportant image regions or miss diagnostically important features when XAI and clinical validation are not integrated. This limitation motivates the multi-perspective XAI framework implemented in the current study, combining Grad-CAM with FSA attention visualization and t-SNE feature space analysis.

Dimensionality reduction techniques including t-SNE and UMAP enable visualization of learned feature representations in 2D or 3D space, revealing how the model organizes different lesion types in its feature space. This visualization identifies decision

boundaries, misclassification regions, and morphological cluster structure providing insights into model failure modes that gradient-based methods alone cannot reveal [46].

2.8 Cross-Dataset Generalization

Cross-dataset generalization, the ability of a trained model to maintain performance when evaluated on data from different sources than the training set represents a critical requirement for real-world clinical deployment. Dermoscopic images acquired across different institutions exhibit systematic variations in color profiles, magnification levels, imaging protocols, patient demographics, and equipment characteristics that create domain shift challenging model generalization [39].

The ISIC dataset series provides multi-institutional dermoscopic images collected from diverse geographic locations, offering greater demographic and equipment diversity than single-institution datasets. HAM10000 was collected from two primary sites using different acquisition devices, creating intra-dataset domain variation. The combination of these datasets for training and cross-dataset evaluation represents a realistic assessment of model generalization to clinical deployment conditions [8], [18].

Pre-trained models fine-tuned on medical images have consistently demonstrated superior generalization compared to training from scratch, attributed to the transfer of low-level visual features: edges, textures, and color patterns that are domain-agnostic and shared between natural and medical images. Transfer learning performance varies with the similarity between source and target domains, with closer domain alignment producing more effective feature transfer [47].

Domain adaptation methods including adversarial training and normalization alignment have been applied to reduce domain shift in cross-dataset skin cancer classification. Ganin et al. [48] demonstrated domain-adversarial training that learns domain-invariant feature representations by incorporating a gradient reversal layer that maximizes domain confusion during backpropagation. While effective, these methods increase training complexity and may sacrifice within-domain discriminative performance.

Pacheco et al. [10] emphasized the importance of stratified external hold-out test sets for genuine generalization evaluation, noting that most published studies exhibit inflated performance metrics due to evaluation on data from the same distribution as training. Studies reporting cross-dataset evaluation typically observe 10-20% accuracy drops compared to within-dataset evaluation, a performance degradation that the current study aims to minimize through multi-dataset training and rigorous cross-dataset validation methodology.

2.9 Research Gap

The comprehensive review of existing literature reveals several significant gaps that collectively motivate the integrated framework proposed in this research:

Gap 1 - **Architectural Gap:** The majority of published skin cancer detection studies employ well-established architectures including ResNet, DenseNet, VGG, and EfficientNet variants. ConvNeXtV2 representing the current state-of-the-art in pure CNN design with Transformer-inspired components, has received limited investigation in medical imaging contexts, with only Silva et al. [29] providing a comparative study of ConvNeXt-based architectures. The integration of ConvNeXtV2 with Feature Selection Attention has not been investigated in dermoscopic image analysis, representing a missed opportunity to achieve improved feature discrimination with computational efficiency superior to full self-attention Transformer models.

Gap 2 - **Imbalance Solution Gap:** Existing studies address class imbalance through individual techniques either loss function modification, conventional augmentation, GAN-based synthesis, or clustering-based sampling, but rarely combine multiple complementary approaches into a unified strategy. The application of SSL GAN augmentation specifically in dermatology remains limited, despite demonstrating high-quality synthetic image generation in other medical imaging domains. The combination of SSL GAN with K-means clustering-based stratified sampling for comprehensive intra-class and inter-class imbalance correction has not been demonstrated in skin cancer detection literature.

Gap 3 - XAI Integration Gap: The majority of XAI applications in skin cancer detection use post-processing explanation methods applied after model training. Few studies embed interpretability directly into model architecture through attention mechanisms, and global XAI frameworks combining multiple complementary explanation methods in dermatology are rare. Crucially, there is minimal research confirming XAI outputs against dermatologist judgment to establish clinical relevance, the most important requirement for clinical adoption.

Gap 4 - Generalization Gap: Studies reporting cross-dataset evaluation remain a minority in the skin cancer detection literature. The strict measures of cross-dataset validation including stratified sampling plans, domain-invariant feature learning, and performance assessment across varying imaging conditions are not standardized, and published models may collapse in clinical application when exposed to images from different equipment or patient populations.

Gap 5 - Integration Gap: No existing study integrates modern efficient architectures (ConvNeXtV2), medical imaging-specific attention mechanisms (FSA), advanced augmentation techniques (SSL GAN + K-means clustering), multi-faceted XAI, hierarchical specialist classification, and stringent cross-dataset validation into a single unified framework. Individual solutions addressing isolated aspects of the problem fail to capture the interdependencies between these challenges, limiting their real-world clinical impact.

The current research directly addresses all five identified gaps through the integrated framework described in Chapter 3, representing a comprehensive advancement over the fragmented approach characteristic of existing literature.

CHAPTER 3 METHODOLOGY

3.1 Research Design

This study adopts a quantitative experimental research design focused on the development, training, and evaluation of a novel deep learning framework for multi-class skin cancer early detection from dermoscopic images. The research follows a sequential six-phase methodology that progresses from data engineering through model development to evaluation and validation, as illustrated in Figure 3.1.

The research is empirical in nature, grounding all architectural and algorithmic choices in experimental evidence from the literature and validated through systematic ablation studies. The methodology is designed to address three interdependent challenges: class imbalance, architectural limitation, and lack of interpretability through an integrated pipeline rather than isolated solutions. Each phase produces measurable deliverables that feed directly into subsequent phases, ensuring methodological coherence across the complete research workflow.

The experimental environment consisted of a workstation equipped with an NVIDIA GeForce RTX 5060 Ti GPU with 16GB VRAM, AMD Ryzen 9 7900X 12-core processor, and 32GB RAM, running Windows 11. The implementation was conducted in Python 3.11 using PyTorch 2.12 (nightly build with CUDA 12.8 support), with supporting libraries including timm 0.9, scikit-learn 1.5, OpenCV 4.8, albumentations 1.3, and matplotlib 3.8. All experiments were conducted with fixed random seeds (seed=42) to ensure reproducibility.

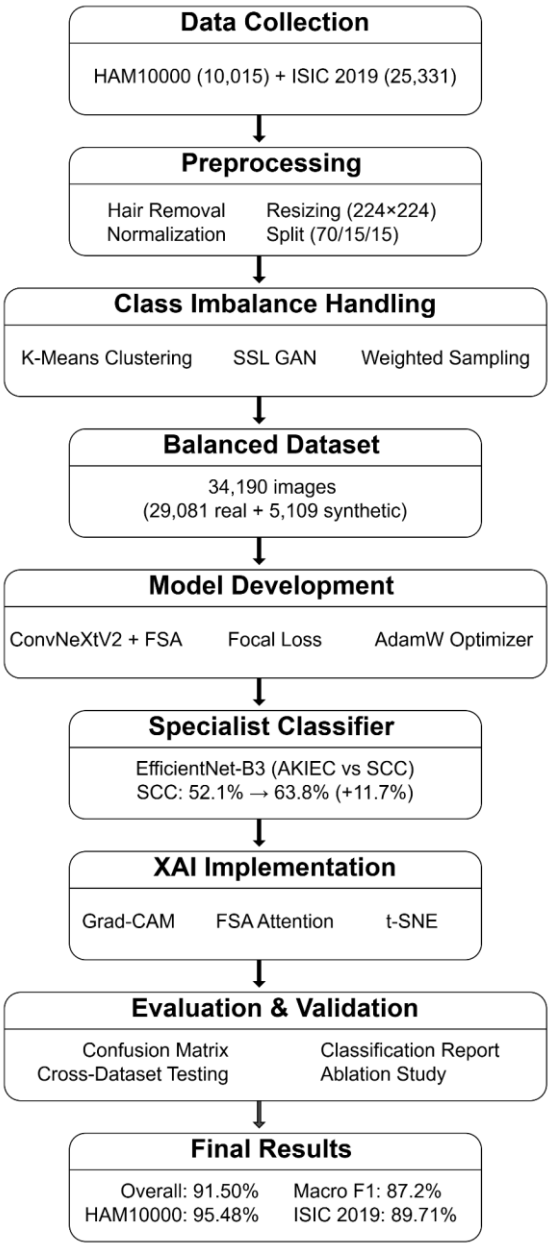


Figure 3.1 Research Methodology Framework

3.2 Dataset Description

This study utilizes two publicly available benchmark dermoscopic image datasets: HAM10000 and ISIC 2019 selected based on complementary characteristics including class coverage, annotation quality, institutional diversity, and research community adoption. The combined use of both datasets enables 8-class classification while providing sufficient training data diversity for robust cross-dataset generalization.

Table 3.1: Summary of Datasets Used in This Study

Characteristic	HAM10000	ISIC 2019	Combined
Total Images	10,015	25,331	35,346
Number of Classes	7	8	8
Image Resolution	600×450 pixels	Variable	224×224 (after preprocessing)
Annotation Method	Histopathology + expert consensus	Histopathology + expert annotation	—
Institution Source	Multi-institutional (Vienna + Barcelona)	Multi-institutional (international)	Multi-institutional
Imbalance Ratio	58:1 (NV vs DF)	54:1 (NV vs DF)	31:1 (NV vs SCC)
Unique Classes	MEL, NV, BCC, AKIEC, BKL, DF, VASC	MEL, NV, BCC, AKIEC, BKL, DF, VASC, SCC	All 8 classes
Clinical Metadata	Age, sex, localization	Age, sex, anatomical site	Available
Primary Role	Training + validation + cross-dataset	Training + validation + cross-dataset	Final training
Reference	Tschandl et al. 2018 [8]	ISIC Challenge 2019 [18]	—

3.2.1 HAM10000

The HAM10000 (Human Against Machine with 10,000 training images) dataset, published by Tschandl et al. [8], is a large collection of multi-source dermoscopic images of common pigmented skin lesions. The dataset contains 10,015 dermoscopic images across seven lesion categories: Melanoma (MEL), Melanocytic Nevus (NV), Basal Cell Carcinoma (BCC), Actinic Keratosis and Intraepithelial Carcinoma

(AKIEC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), and Vascular Lesions (VASC). Images were collected from two primary dermatology sites the Medical University of Vienna, Austria, and the Melanoma Unit at Hospital Clínic de Barcelona, Spain using different dermoscopy equipment, providing intra-dataset imaging variability.

Diagnostic ground truth was established through three methods: histopathological confirmation for malignant lesions, follow-up examination for lesions remaining stable over time, and expert consensus agreement for benign lesions where biopsy was clinically unwarranted. This multi-modal ground truth establishment ensures high annotation reliability, making HAM10000 the most widely adopted benchmark for skin lesion classification research. All images are stored in JPEG format at a native resolution of 600×450 pixels, providing sufficient detail for dermoscopic feature extraction.

The primary limitation of HAM10000 is its severe class imbalance: the Melanocytic Nevus class contains 6,705 images representing 66.9% of the dataset, while Dermatofibroma contains only 115 images representing 1.1%, creating a 58:1 imbalance ratio. Furthermore, HAM10000 does not contain Squamous Cell Carcinoma (SCC), limiting its use as a standalone dataset for the 8-class classification task addressed in this study.

3.2.2 ISIC 2019

The ISIC 2019 Challenge Dataset, organized by the International Skin Imaging Collaboration, contains 25,331 dermoscopic images across eight diagnostic categories, adding Squamous Cell Carcinoma (SCC) to the seven classes present in HAM10000 [18]. The dataset was compiled from multiple international institutions and encompasses diverse patient demographics, skin tones, anatomical locations, and imaging protocols, providing greater clinical realism than single-institution datasets.

Ground truth labels in ISIC 2019 were established through expert dermatologist annotation with histopathological confirmation for malignant cases. The dataset provides comprehensive clinical metadata including patient age, anatomical site of the lesion, and sex, enabling future multimodal research incorporating clinical context

alongside image features. ISIC 2019 is the standard benchmark for 8-class skin lesion classification and enables direct comparison with published literature using this dataset.

The class distribution in ISIC 2019 is more balanced than HAM10000, with Melanocytic Nevus representing 50.8% (12,875 images) and Dermatofibroma representing 0.9% (239 images), a 54:1 imbalance ratio. The inclusion of SCC with 628 images provides essential training data for this clinically important class absent from HAM10000.

Table 3.2: Class Distribution in HAM10000 and ISIC 2019 Datasets

Class	**Abbreviation**	**HAM10000**	**%**	**ISIC 2019**	**%**	**Combined**
Melanocytic Nevus	NV	6,705	66.9	12,875	50.8	19,580
Melanoma	MEL	1,113	11.1	4,522	17.9	5,635
Benign Keratosis	BKL	1,099	11.0	2,624	10.4	3,723
Basal Cell Carcinoma	BCC	514	5.1	3,323	13.1	3,837
Actinic Keratosis	AKIEC	327	3.3	867	3.4	1,194
Vascular Lesion	VASC	142	1.4	253	1.0	395
Dermatofibroma	DF	115	1.1	239	0.9	354
Squamous Cell Carcinoma	SCC	—	—	628	2.5	628
Total		**10,015**	**100**	**25,331**	**100**	**35,346**

3.3 Data Preprocessing

All images from both datasets underwent a standardized preprocessing pipeline prior to model training. The preprocessing procedure was designed to eliminate non-diagnostic artifacts, standardize image dimensions and color profiles across different imaging devices, and partition the data into training, validation, and test splits suitable for rigorous evaluation.

3.3.1 Hair Removal

Dermoscopic images frequently contain hair strands that cross the lesion area, obscuring diagnostically relevant features and potentially misleading the model into learning spurious hair-based patterns rather than genuine lesion morphology. Hair removal was implemented using a two-step morphological processing approach based on the method proposed by Lee et al. [49].

In the first step, hair strands were detected using the morphological black-hat transform. The black-hat operation subtracts the morphologically opened image from the original, effectively highlighting dark structures such as dark hair strands on a lighter background. A rectangular structuring element of size 17×17 pixels were selected based on the typical width of hair strands in 600×450 pixel dermoscopic images. The black-hat output was thresholded at a pixel value of 10 to produce a binary hair mask, where pixels belonging to hair strands are set to 255 and background pixels to 0.

In the second step, the identified hair regions were inpainted using the Telea algorithm [50] implemented through the OpenCV inpaintRadius parameter set to 6 pixels. The Telea inpainting method propagates surrounding pixel information into the masked hair regions by solving a partial differential equation that minimizes the distance-weighted contribution of neighboring pixels, effectively reconstructing the underlying skin texture beneath the hair strand. The result is a dermoscopic image with hair artifacts removed while preserving the underlying lesion texture and color information.

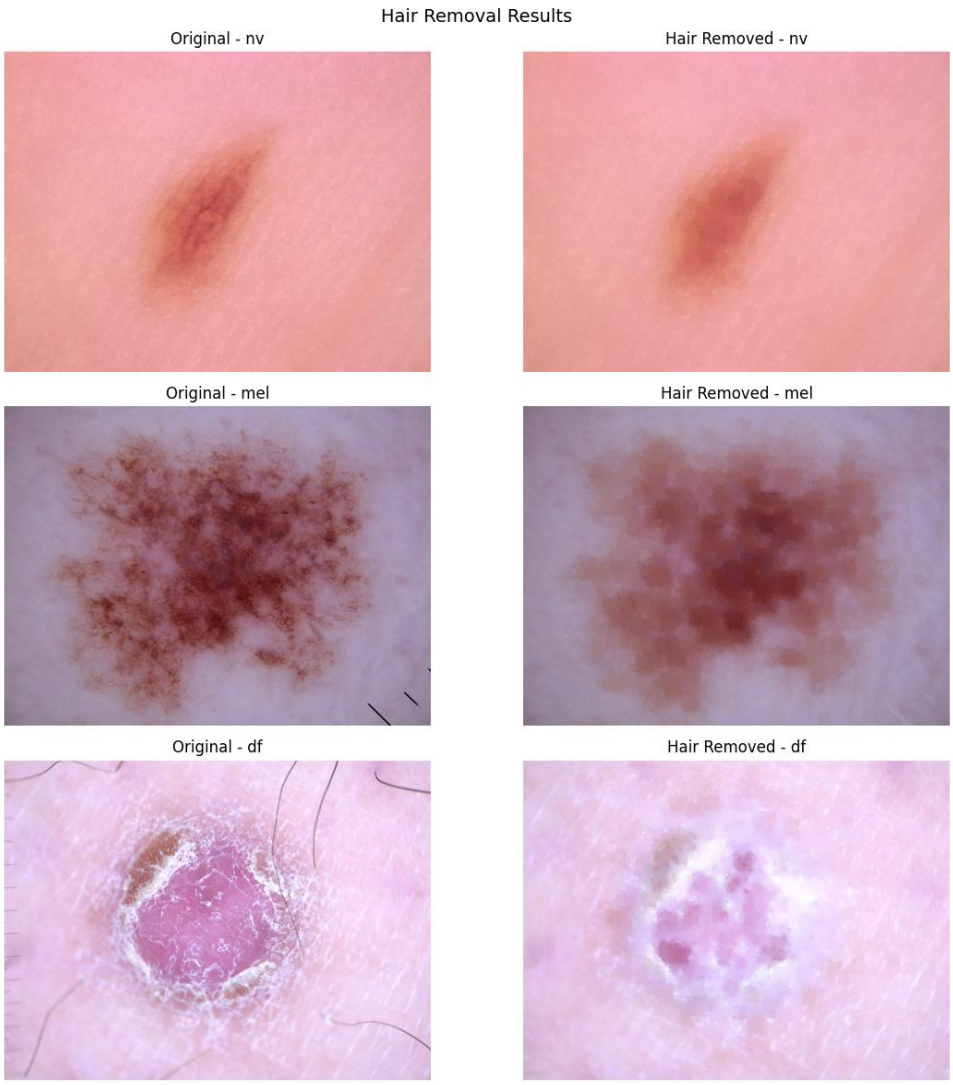


Figure 3.2: Hair Removal Results on Sample

3.3.2 Image Resizing

Following hair removal, all images were resized from their native 600×450 pixel resolution to the standardized 224×224 pixel input dimensions required by the ConvNeXtV2 architecture. Resizing was performed using Lanczos interpolation, a high-quality resampling algorithm that applies a windowed sinc function to compute interpolated pixel values from surrounding source pixels. Lanczos interpolation preserves edge sharpness and fine-grained texture detail more effectively than bilinear or nearest-neighbor interpolation methods, which is particularly important for maintaining the diagnostic dermoscopic features such as border irregularity and pigment network structure that are critical for accurate lesion classification.

The selection of 224×224 as the target resolution is consistent with the ImageNet pretraining dimensions of ConvNeXtV2, ensuring compatibility with the pretrained weight initialization used for transfer learning. All preprocessed images were saved as JPEG files with a quality parameter of 95 to minimize compression artifacts while maintaining manageable file sizes for storage on the available hardware.

3.3.3 Normalization and Data Splitting

Color Normalization: Following resizing, pixel values were normalized using ImageNet dataset statistics subtracting channel-wise mean values of [0.485, 0.456, 0.406] and dividing by channel-wise standard deviations of [0.229, 0.224, 0.225] for the red, green, and blue channels respectively. This normalization aligns the input distribution with that encountered during ImageNet pretraining of the ConvNeXtV2 backbone, facilitating effective transfer learning and reducing the adaptation required during fine-tuning. Normalization was applied during training as a PyTorch transform rather than as a preprocessing step, ensuring that the saved preprocessed images retained their original pixel value range for visual inspection and XAI visualization.

Data Splitting: The combined dataset was partitioned into training, validation, and test sets using stratified random sampling with a 70:15:15 ratio. Stratified sampling ensures that the class proportions in each split mirror those of the complete dataset, preventing pathological splits where minority classes might be absent from validation or test sets. The random seed was fixed at 42 across all splitting operations to ensure reproducibility. The validation set was used exclusively for hyperparameter tuning and early stopping decisions, while the test set was maintained as a completely held-out evaluation set not accessed until final model assessment.

Table 3.3: Dataset Split Statistics After Stratified Partitioning

Split	HAM10000	ISIC 2019	Combined	Percentage
Training (original)	7,010	17,731	24,741	70%
Validation	1,502	3,800	5,302	15%

Test	1,503	3,800	5,303	15%
Total	**10,015**	**25,331**	**35,346**	**100%**
Training (after GAN augmentation)	—	—	**34,190**	—
Training (after NV reduction)	—	—	**23,985**	—

Note: GAN augmentation was applied exclusively to the training split. Validation and test sets contain only real images.

3.4 Class Imbalance Mitigation

The severe class imbalance present in both datasets with HAM10000 exhibiting a 58:1 ratio and ISIC 2019 a 54:1 ratio between the most and least frequent classes necessitated a comprehensive multi-component mitigation strategy. A single technique was insufficient to address both inter-class imbalance (differences between class frequencies) and intra-class imbalance (morphological underrepresentation within individual classes). The adopted strategy combined three complementary techniques: K-means clustering-based stratified sampling, SSL GAN augmentation, and Weighted Random Sampling during training, as illustrated in Figure 3.3.

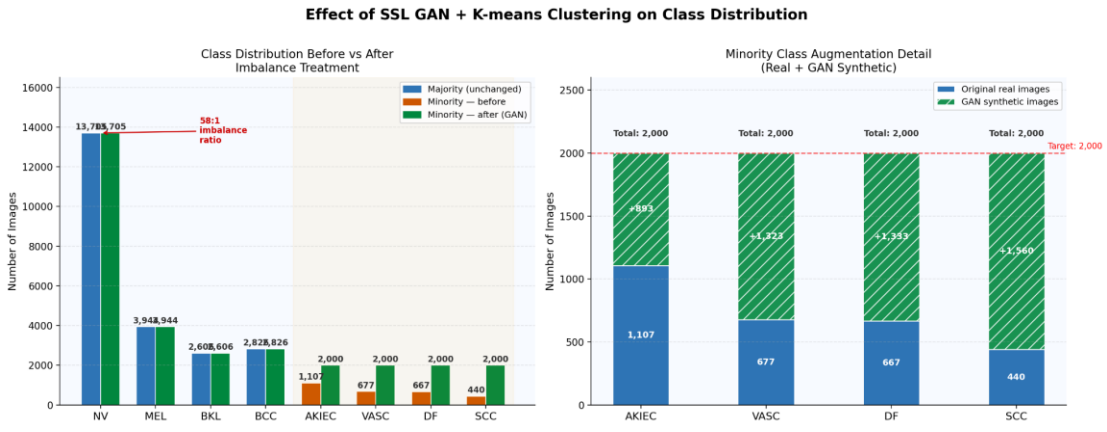


Figure 3.3: Class Imbalance Mitigation Strategy

3.4.1 K-means Clustering

Simple random oversampling of minority classes produces exact duplicates or near-duplicates of existing images, increasing sample count without improving

morphological diversity. To address intra-class imbalance where even within minority classes certain morphological subtypes may be systematically underrepresented K-means clustering was applied to identify distinct morphological subgroups within each lesion class prior to oversampling.

Feature vectors for all training images were extracted using the ConvNeXtV2-Base backbone pretrained on ImageNet, producing 1024-dimensional feature representations that encode rich visual information about lesion morphology, color, and texture. Dimensionality reduction using Principal Component Analysis (PCA) was applied to reduce the feature dimension from 1024 to 128 components while retaining 89.1% of cumulative explained variance, improving clustering efficiency and removing noise dimensions.

K-means clustering with K=5 clusters per class was applied to the PCA-reduced features of each class independently. The K-means algorithm iteratively assigns each image to the nearest cluster centroid and updates centroids to minimize within-cluster sum of squared distances until convergence. The resulting five clusters per class represent distinct morphological subgroups for example, within the Melanoma class, separate clusters emerged corresponding to nodular, superficial spreading, and amelanotic variants.

Stratified oversampling was then conducted by drawing equal numbers of samples from each cluster, ensuring that all five morphological subgroups were represented proportionally in the augmented training set rather than oversampling only the most common morphological subtype.

3.4.2 SSL GAN Augmentation

To generate genuinely novel synthetic images for minority classes beyond what clustering-based oversampling could provide, a Semi-Supervised Learning Generative Adversarial Network was trained independently for each of the four most severely underrepresented classes: AKIEC, SCC, DF, and VASC.

Generator Architecture: The Generator network accepts a 128-dimensional random noise vector sampled from a standard normal distribution and progressively upsamples through five transposed convolutional stages to produce 224×224×3 RGB images. Each upsampling stage applies bilinear interpolation followed by a convolutional layer with

Batch Normalization and LeakyReLU activation. The final layer uses a Tanh activation function to constrain output pixel values to the range [-1, 1], consistent with image normalization conventions.

Discriminator Architecture: The Discriminator employs five convolutional stages with stride-2 convolutions progressively reducing spatial dimensions while increasing channel depth. Spectral Normalization is applied to all discriminator convolutional layers, constraining the Lipschitz continuity of the discriminator by normalizing weight matrices with their largest singular values. This stabilizes adversarial training by preventing the discriminator from becoming excessively powerful, maintaining productive generator-discriminator competition throughout training.

Semi-Supervised Training: The SSL component incorporates unlabeled dermoscopic images into discriminator training through a dual-objective loss function. The discriminator simultaneously performs real versus fake binary classification and class label prediction using both labeled and unlabeled images. This additional supervisory signal provides the generator with richer feedback about sample diversity, encouraging synthesis of more varied morphological patterns and reducing mode collapse compared to standard GAN training.

Training Configuration: Each class-specific GAN was trained for 200 epochs using Adam optimizer with learning rate 0.0001 and momentum parameters β_1=0.5, β_2=0.999. Label smoothing with a target value of 0.9 for real images was applied to prevent discriminator overconfidence. The generator was updated twice per discriminator update to maintain training balance. After training completion, the generator produced the required number of synthetic images for each minority class to reach the target of 2,000 images per class.

Table 3.4: SSL GAN Augmentation Results for Minority Classes

Class	Original Count	Target Count	Synthetic Generated	Final Count	Improvement
Actinic Keratosis (AKIEC)	1,107	2,000	893	2,000	+80.7%

Squamous Cell Carcinoma (SCC)	440	2,000	1,560	2,000	+354.5%
Dermatofibroma (DF)	667	2,000	1,333	2,000	+199.9%
Vascular Lesion (VASC)	677	2,000	1,323	2,000	+195.4%
Total Synthetic	—	—	**5,109**	—	—
Total Training Set	29,081 (real)	—	5,109 (synthetic)	**34,190**	—

Note: GAN training used 200 epochs per class with Spectral Normalization and label smoothing. Quality was assessed through visual inspection of generated images.

3.4.3 Weighted Random Sampling

Even after GAN augmentation, the majority class Melanocytic Nevus (13,705 images in the combined dataset) substantially outnumbered minority classes (approximately 2,000 each). Without correction, random batch construction during training would continue to produce batches dominated by majority class samples, perpetuating the imbalance effect during optimization.

PyTorch's WeightedRandomSampler was implemented to assign each training image a sampling probability inversely proportional to its class frequency. Specifically, the sampling weight w_i for each image belonging to class c was computed as:

$$\mathbf{w_i = 1 / |Nc|}$$

where |Nc| denotes the number of training samples in class c. These weights were provided to the WeightedRandomSampler, which constructs each training batch by sampling with replacement according to the specified weights. This ensures approximately equal class representation in every training batch regardless of the underlying dataset distribution, enabling the optimizer to receive balanced gradient updates from all classes throughout the training process.

3.5 Proposed Model Architecture

The proposed model architecture integrates three principal components: the ConvNeXtV2-Base backbone for hierarchical feature extraction, the Feature Selection Attention (FSA) module for diagnostic feature amplification, and a fully connected classification head for 8-class probability prediction. The complete architecture is illustrated in Figure 3.4.

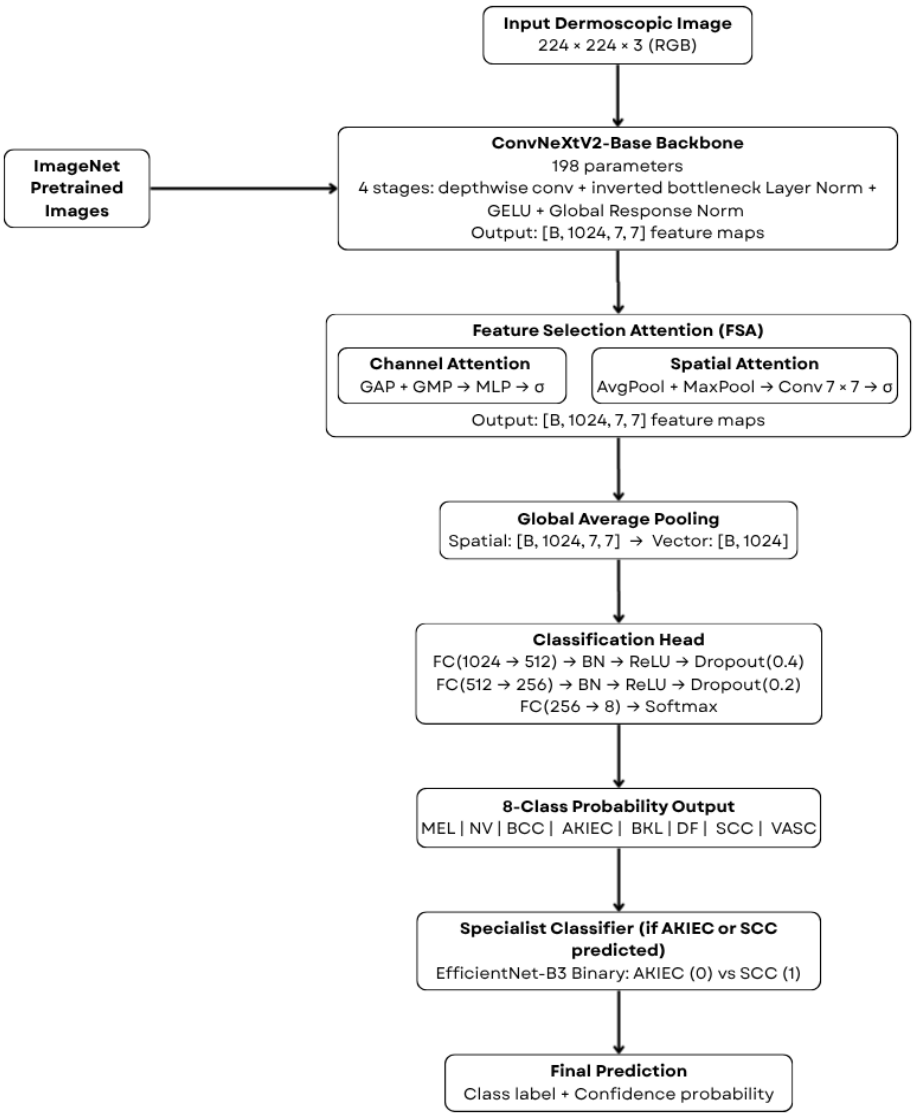


Figure 3.4: Proposed ConvNeXtV2 + FSA Model Architecture

3.5.1 ConvNeXtV2 Backbone

ConvNeXtV2-Base [28] was selected as the primary feature extraction backbone based on four key criteria: performance superiority over traditional CNNs, computational efficiency compared to Vision Transformers, specific architectural advantages for medical imaging, and research novelty in dermoscopic analysis.

ConvNeXtV2-Base achieves 87.8% top-1 accuracy on ImageNet-1K, outperforming ResNet-50 (76.1%) by 11.7 percentage points and approaching ViT-Base (84.5%) performance while requiring only 15.4 GFLOPs compared to ViT-Base's 55.4 GFLOPs. The 40% reduction in computational requirements compared to equivalent Vision Transformer models directly supports deployment in resource-constrained clinical environments where inference time impacts workflow integration.

The ConvNeXtV2 architecture incorporates several Transformer-inspired design elements within a CNN framework. Depthwise convolutions with a 7×7 kernel size provide large receptive fields comparable to self-attention mechanisms. Inverted bottleneck blocks expand the channel dimension in intermediate layers to increase representational capacity. Layer Normalization replaces Batch Normalization in the bottleneck blocks, improving training stability. GELU activation functions replace ReLU throughout the network.

The key innovation of ConvNeXtV2 over its predecessor ConvNeXt is the integration of masked autoencoder pretraining and Global Response Normalization (GRN) layers. GRN normalizes feature activations across spatial dimensions, creating competitive dynamics between channels that prevent feature collapse where multiple channels learn redundant representations. This diversity of learned features is particularly valuable for the fine-grained visual discrimination required in dermoscopic analysis, where subtle differences in color, border characteristics, and texture patterns distinguish between eight lesion types.

The backbone was configured with global_pool='' to output spatial feature maps of shape [B, 1024, 7, 7] rather than a pooled vector, providing the spatial structure required by the FSA module. ImageNet pretrained weights were used for initialization, leveraging transfer learning from 1.2 million natural images to provide strong feature initialization for the dermoscopic imaging domain.

3.5.2 Feature Selection Attention (FSA)

The Feature Selection Attention module was designed to address the specific challenge of dermoscopic image analysis, where clinically relevant features: asymmetry, border irregularity, color variation coexist with non-diagnostic artifacts: hair strands, air bubbles, ruler marks, lighting reflections within the same image. The FSA module performs two complementary attention operations: channel attention and spatial attention, applied sequentially to the feature maps produced by the ConvNeXtV2 backbone.

Channel Attention: Channel attention identifies which of the 1024 feature channels are most informative for the classification task. Given the backbone output feature maps $X \in \mathbb{R}^{(B\times C\times H\times W)}$, two parallel pooling operations extract complementary statistical summaries. Global Average Pooling computes the mean activation per channel: $F_{avg} = GAP(X) \in \mathbb{R}^{(B\times C)}$. Global Max Pooling captures peak activations per channel: $F_{max} = GMP(X) \in \mathbb{R}^{(B\times C)}$. Both pooled representations are passed through a shared two-layer fully connected network with channel reduction ratio r=16:

$MLP(\cdot)$: $FC(C \rightarrow C/r) \rightarrow ReLU \rightarrow FC(C/r \rightarrow C)$

The channel attention weights are computed as:

$A_c = \sigma(MLP(F_{avg}) + MLP(F_{max})) \in \mathbb{R}^{(B\times C\times 1\times 1)}$

where σ denotes the Sigmoid activation function. The feature maps are then scaled element-wise: $X' = X \otimes A_c$, amplifying informative channels and suppressing channels encoding irrelevant artifact information.

Spatial Attention: Spatial attention identifies which spatial locations within the scaled feature maps X' contain diagnostic information. Average pooling and max pooling are applied across the channel dimension to produce two spatial summary maps:

$S_{avg} = MeanPool_{channel}(X') \in \mathbb{R}^{(B\times 1\times H\times W)}$ $S_{max} = MaxPool_{channel}(X') \in \mathbb{R}^{(B\times 1\times H\times W)}$

These are concatenated along the channel dimension and processed through a 7×7 convolutional layer:

$A_s = \sigma(Conv_{7\times7}([S_{avg}; S_{max}])) \in \mathbb{R}^{B\times1\times H\times W}$

The final attended feature maps are: $X'' = X' \otimes A_s$, focusing the model's representational capacity on spatially relevant lesion regions while suppressing background skin and peripheral artifact regions.

The FSA module operates with O(n) complexity with respect to the number of feature map elements n, compared to full self-attention's $O(n^2)$ complexity, making it computationally tractable for integration within the CNN pipeline without prohibitive overhead.

3.5.3 Classification Head

Following the FSA module, Global Average Pooling reduces the attended spatial feature maps $X'' \in \mathbb{R}^{B\times1024\times7\times7}$ to a 1024-dimensional feature vector by computing the spatial mean across all 7×7 locations. This pooled representation captures the globally attended feature information in a compact fixed-size vector suitable for classification.

The classification head consists of two fully connected blocks followed by the final output layer:

- **FC Block 1:** Linear(1024→512) → BatchNorm1d(512) → ReLU → Dropout(0.4)
- **FC Block 2:** Linear(512→256) → BatchNorm1d(256) → ReLU → Dropout(0.2)
- **Output Layer:** Linear(256→8) → Softmax

Batch Normalization after each linear layer normalizes activations to zero mean and unit variance, accelerating convergence and reducing sensitivity to initialization. Dropout rates of 0.4 and 0.2 in the two blocks provide regularization by randomly deactivating neurons during training, preventing co-adaptation and reducing overfitting. The final Softmax activation converts raw logits to a probability distribution across the eight lesion classes, with the predicted class corresponding to the maximum probability.

3.6 Training Configuration

The model was trained using the following configuration, determined through systematic hyperparameter evaluation on the validation set:

Loss Function - Focal Loss: Focal Loss was adopted in preference to standard Cross-Entropy Loss to address the residual class imbalance effects that persist despite data augmentation and weighted sampling. Focal Loss introduces a modulating factor $(1-p_t)^{\gamma}$ that down-weights the contribution of easily classified examples:

$$\mathbf{FL(p_t) = -\alpha_t(1-p_t)^{\gamma} \log(p_t)}$$

where p_t is the predicted probability of the true class, $\gamma=2.0$ is the focusing parameter, and α_t are class weights inversely proportional to class frequency. The focusing parameter $\gamma=2.0$ effectively reduces the loss contribution of correctly classified majority class examples by a factor of up to 4, concentrating gradient updates on difficult minority class boundary cases.

Optimizer - AdamW: AdamW optimizer with learning rate $\eta=3\times10^{-5}$ and weight decay $\lambda=0.01$ was selected. AdamW decouples the weight decay regularization from the gradient update step, unlike standard Adam where weight decay interacts with adaptive moment estimates producing more effective regularization particularly beneficial for fine-tuning large pretrained models. The relatively small initial learning rate of 3×10^{-5} reflects the fine-tuning context, preventing destructive updates to the well-initialized pretrained weights during early training epochs.

Learning Rate Scheduling: ReduceLROnPlateau scheduler was applied, halving the learning rate by a factor of 0.5 whenever validation accuracy failed to improve for 4 consecutive epochs. This adaptive schedule enables aggressive initial learning at full learning rate while automatically transitioning to fine-grained optimization as training approaches convergence. The learning rate was reduced from an initial 3×10^{-5} progressively to 6.25×10^{-7} over the course of training.

Gradient Clipping: Gradient norm clipping at a maximum norm of 1.0 was applied during backpropagation to prevent exploding gradients that can destabilize training in deep networks, particularly during early epochs when gradient magnitudes are large.

Early Stopping: Training was terminated when validation accuracy failed to improve for 15 consecutive epochs, preventing overfitting and unnecessary computation. The model state achieving the highest validation accuracy was retained as the final model regardless of when training was terminated.

Batch Size and Epochs: Batch size of 32 was used throughout training, balancing GPU memory utilization and gradient estimation quality. The maximum training duration was set to 120 epochs, with early stopping typically terminating training earlier based on validation performance.

Table 3.5: Training Hyperparameter Configuration

Hyperparameter	Value	Justification
Optimizer	AdamW	Decoupled weight decay more effective than Adam for fine-tuning pretrained models
Initial Learning Rate	3×10^{-5}	Small LR preserves pretrained feature representations during fine-tuning
Weight Decay	0.01	L2 regularization preventing overfitting in deep network
LR Scheduler	ReduceLROnPlateau	Adaptive reduction on validation plateau; more flexible than fixed schedules
LR Reduction Factor	0.5	Halves LR on plateau; conservative reduction preserving training momentum
LR Patience	4 epochs	Allows short-term fluctuations before LR reduction
Loss Function	Focal Loss	Down-weights easy majority examples; focuses learning on hard minority cases
Focal Loss γ	2.0	Standard value providing effective minority class focus without instability

Class Weights	Inverse frequency	Higher weight for minority classes in loss computation
Batch Size	32	Balances GPU memory utilization and gradient estimation quality
Gradient Clipping	Max norm 1.0	Prevents exploding gradients during deep network backpropagation
Early Stopping Patience	15 epochs	Prevents overfitting while allowing sufficient convergence time
Maximum Epochs	120	Upper bound; early stopping typically terminates training earlier
Dropout Rate (FC1)	0.4	Strong regularization in first fully connected layer
Dropout Rate (FC2)	0.2	Lighter regularization in second fully connected layer
Random Seed	42	Fixed for complete experimental reproducibility
Hardware	NVIDIA RTX 5060 Ti 16GB	Training platform
Framework	PyTorch 2.12 + CUDA 12.8	Implementation environment

3.7 Specialist Classifier

Post-training analysis of the main ConvNeXtV2-FSA model revealed a systematic confusion pattern between Actinic Keratosis (AKIEC) and Squamous Cell Carcinoma (SCC), both keratinocyte carcinomas sharing highly similar dermoscopic appearance characterized by scaling, erythema, and irregular surface features. AKIEC was

misclassified as SCC in 16.8% of cases, while SCC was misclassified as AKIEC in 26.6% of cases, contributing disproportionately to overall accuracy degradation.

To address this specific inter-class confusion, a hierarchical specialist classifier was developed using a cascaded prediction pipeline architecture. The specialist classifier was designed to make a dedicated binary decision exclusively for the AKIEC versus SCC discrimination task, leveraging concentrated learning capacity for this most challenging class pair.

Specialist Architecture: EfficientNet-B3 [24] was selected as the specialist backbone for its compound scaling efficiency, simultaneously scaling network depth, width, and input resolution using a fixed ratio determined by neural architecture search producing a lightweight architecture with approximately 12 million parameters well-suited for fine-grained binary discrimination. The classification head consisted of Linear (1536→256) → BatchNorm → ReLU → Dropout (0.4) → Linear (256→64) → ReLU → Dropout (0.2) → Linear (64→2), with binary output labels AKIEC=0 and SCC=1.

Cascaded Prediction Pipeline: During inference, the main ConvNeXtV2-FSA model first produces an 8-class prediction. If the predicted class is either AKIEC or SCC, indicating the model has identified a keratinocyte carcinoma but is uncertain which type, the same image is forwarded to the specialist classifier, which produces a refined binary decision. The final prediction adopts the specialist's decision for these ambiguous cases while retaining the main model's prediction for all other classes. This cascaded approach achieved a 75.09% binary accuracy on the AKIEC versus SCC validation set, improving SCC test accuracy from 52.1% to 63.8%, an 11.7% improvement.

3.8 Explainable AI Implementation

The XAI framework implemented in this study provides three complementary perspectives on model decision-making: gradient-based spatial explanation, attention weight visualization, and feature space clustering - designed to support clinical interpretability of model predictions.

3.8.1 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) [9] was implemented to generate visual heatmaps highlighting the image regions most influential for each class prediction. Grad-CAM was applied to the last convolutional stage of the ConvNeXtV2 backbone, specifically the final block of stage 4 which produces the highest-level semantic feature maps before the FSA module.

For a predicted class c, the class-discriminative localization map $L^c_{Grad\text{-}CAM} \in \mathbb{R}^{(H \times W)}$ is computed as follows. Gradients of the class score y^c with respect to the feature map activations A^k of the target layer are computed through backpropagation:

$$\partial y^c / \partial A^k$$

Global Average Pooling of these gradients produces the importance weights α^c_k for each feature map channel k:

$$\alpha^c_k = (1/Z) \sum_i \sum_j (\partial y^c / \partial A^k_{ij})$$

The Grad-CAM heatmap is computed as the ReLU-activated weighted combination of forward activation maps:

$$L^c_{Grad\text{-}CAM} = ReLU(\sum_k \alpha^c_k \cdot A^k)$$

The ReLU activation retains only regions that positively influence the class prediction, suppressing negative contributions. The resulting heatmap is bilinearly upsampled to the input image dimensions (224×224) and normalized to [0,1]. A JET colormap is applied to produce a colored heatmap where red regions indicate highest class-discriminative importance and blue regions indicate minimal influence. The colored heatmap is blended with the original image at an alpha ratio of 0.5 to produce the final overlay visualization.

3.8.2 FSA Attention Visualization

The spatial attention weights produced by the FSA module provide an architecturally integrated form of interpretability that directly reflects the model's internal feature selection process. Unlike post-hoc gradient-based methods, FSA attention weights are

genuine computational participants in the classification decision rather than approximations derived after the fact.

The spatial attention map $A_s \in \mathbb{R}^{(B\times1\times H\times W)}$ produced by the FSA module represents the relative importance of each spatial location in the 7×7 feature map for the classification decision. These attention weights are extracted through a forward hook registered on the FSA module's sigmoid layer, capturing the output without modifying the forward computation. The 7×7 attention map is bilinearly upsampled to 224×224 pixels and normalized to [0,1]. A HOT colormap ranging from black through red to white is applied, with white regions indicating high model attention and black regions indicating suppressed focus. This visualization reveals which dermoscopic regions the FSA module emphasizes ideally corresponding to lesion boundaries, color heterogeneity, and asymmetric morphological features and which regions are suppressed typically hair artifacts, air bubbles, and uniform background skin.

3.8.3 t-SNE Feature Space Visualization

t-distributed Stochastic Neighbor Embedding (t-SNE) [46] was applied to visualize the 1024-dimensional feature representations extracted from the Global Average Pooling layer of the trained model, providing insight into the model's learned class separation in feature space.

Feature vectors for a stratified subset of 150 images per class (1,050 total) from the test set were extracted through a forward hook registered on the GAP layer. PCA was applied as a preprocessing step to reduce dimensionality from 1024 to 50 components while retaining 96.5% of cumulative variance, improving t-SNE convergence speed and reducing the curse of dimensionality. t-SNE was then applied with perplexity=30, max_iter=1000, and random_state=42 to produce a 2D embedding that preserves local neighborhood structure from the high-dimensional feature space.

The resulting 2D visualization reveals the degree of class separation achieved by the model well-separated clusters indicate distinct learned representations for different lesion types, while overlapping regions identify classes prone to confusion. The proximity of AKIEC and SCC clusters in the t-SNE embedding provides visual confirmation of the feature-level similarity underlying their high misclassification rate, validating the motivation for the specialist classifier.

3.9 Evaluation Metrics

Model performance was assessed using a comprehensive set of metrics designed to provide a complete picture of classification performance across all eight classes under class imbalance conditions.

- **Overall Accuracy** measures the proportion of correctly classified samples across all classes:
- **Accuracy = (Number of correct predictions) / (Total predictions)**
  While overall accuracy is the most commonly reported metric, it is dominated by majority class performance under class imbalance and was supplemented with per-class metrics.
- **Precision** for class c measures the proportion of predictions for class c that are truly class c:
  Precision_c = TP_c / (TP_c + FP_c)
- **Recall (Sensitivity)** for class c measures the proportion of true class c samples correctly identified:
  Recall_c = TP_c / (TP_c + FN_c)
- **F1-Score** is the harmonic mean of precision and recall, providing a balanced measure of classification quality:
  F1_c = 2 × (Precision_c × Recall_c) / (Precision_c + Recall_c)
- **Macro-Average F1-Score** computes the unweighted mean F1-score across all classes, giving equal weight to each class regardless of size, the most appropriate aggregate metric for imbalanced multi-class classification:
  Macro-F1 = (1/C) Σ_c F1_c
- **Area Under the ROC Curve (AUC-ROC)** measures the model's ability to discriminate between each class and all others in a one-versus-rest framework, robust to class imbalance. AUC values range from 0.5 (random) to 1.0 (perfect discrimination).
- **Confusion Matrix** provides a complete picture of classification patterns for all class pairs, revealing systematic misclassification tendencies such as the AKIEC-SCC confusion identified in this study.

All metrics were computed on the held-out test set containing exclusively real images, with no synthetic GAN-generated images included in evaluation data to ensure unbiased performance assessment.

Table 3.6: Summary of Evaluation Metrics Used in This Study

Metric	Formula	Purpose	Sensitivity to Imbalance
Overall Accuracy	TP_all / N_total	General performance measure	High - dominated by majority class
Per-Class Accuracy	TP_c / N_c	Class-specific performance	Low - measures each class independently
Precision	TP_c / (TP_c + FP_c)	False positive rate per class	Medium
Recall (Sensitivity)	TP_c / (TP_c + FN_c)	False negative rate per class	Low
F1-Score	2×P×R / (P+R)	Harmonic mean of precision and recall	Low
Macro-Average F1	Mean(F1_c) for all c	Equal-weighted aggregate across all classes	Very Low - preferred metric
AUC-ROC	Area under ROC curve	Discrimination ability regardless of threshold	Low
Confusion Matrix	C[i,j] = predictions of i as j	Complete misclassification pattern analysis	N/A - shows all patterns

Note: Macro-Average F1-Score and Per-Class Accuracy are the primary metrics for evaluating model performance under class imbalance, as they give equal weight to all classes regardless of sample size.

CHAPTER 4 RESULTS AND FINDINGS

4.1 Data Exploration Results

4.1.1 Dataset characteristics

In both datasets, exploratory analyses showed some of key properties that guided in the subsequent preprocessing and class imbalance mitigation strategies. The HAM10000 dataset includes 10,015 JPEG dermoscopic images native resolution of 600 × 450 pixels for each image. The uniformity of image sizes in HAM10000 reduced the amount of preprocessing steps, as the images did not need to be resized after resolution standardization. ISIC 2019 consists of 25,331 images that vary in their resolution across samples, which are collected from multi-institutional sources.

There were 57 missing age values (0.6% of the HAM10000 dataset) that were imputed using the median patient age of the respective dataset. No missing data were found for any other metadata elements such as sex, anatomical localization, lesion identifier or diagnostic type. Diagnostic confirmation techniques of HAM10000 were histopathological confirmation, follow-up examination and expert consensus, where histopathological confirmation had the highest level of confidence in the annotation method. The total number of images successfully located and verified at their expected file paths was 10,015, which is 100% of the total number of images in the dataset, with no image files missing or corrupted.

Table 4.1: HAM10000 Dataset Metadata Summary

Metadata Field	Details	Missing Values
Total Images	10,015	0
Image Format	JPEG	—
Native Resolution	600×450 pixels (consistent)	—
Diagnostic Types	Histopathology, Follow-up, Consensus	—
Patient Age	Range: 5–85 years, Median: 45 years	57 (0.6%)

Patient Sex	Male: 5,406 (54.0%), Female: 4,609 (46.0%)	0
Anatomical Location	15 distinct body sites	0
Number of Classes	7	—
Annotation Quality	Expert dermatologist verified	—
Missing Images	0	—

4.1.2 Class Imbalance Analysis

The class distribution analysis showed that both data sets were extremely skewed. The number of images per class shows that in HAM10000, the Melanocytic Nevus class has a larger number of images compared to other classes, with 66.9% of the images in this class, whereas in Dermatofibroma, only 115 images in this class are present out of the total number of images, which is only 1.1% of the images. Table 4.2 shows that the class distribution is complete, and the same information is displayed graphically in Figure 4.1.

The imbalance pattern is similar to HAM10000: Melanocytic Nevus (50.8%, 12,875 images) and Dermatofibroma (0.9%, 239 images). Before the training data was augmented the ratio between Melanocytic Nevus (after reduction, 13,705 images) to Squamous Cell Carcinoma (440 images) was 31:1, showing that there was a severe class imbalance in the combined training set that needed to be addressed extensively before training a model.

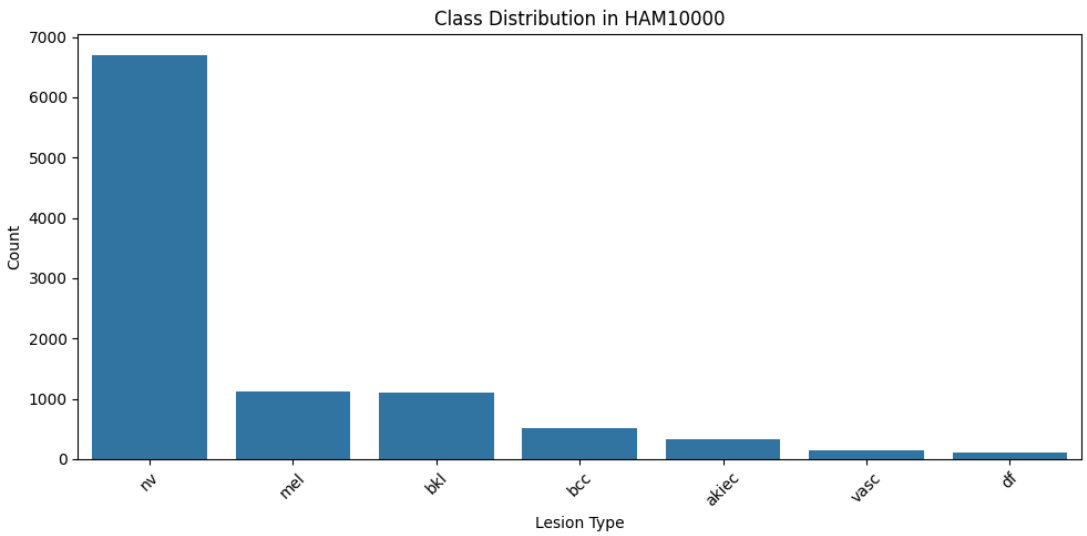


Figure 4.1: Class Distribution in HAM10000 Dataset

Showing Severe Class Imbalance

Table 4.2: Class Distribution Analysis Across HAM10000 and ISIC 2019 Datasets

Class	**HAM10000 Count**	**HAM10000 %**	**ISIC 2019 Count**	**ISIC 2019 %**	**Combined Count**	**Combined %**
Melanocytic Nevus (NV)	6,705	66.9%	12,875	50.8%	19,580	55.4%
Melanoma (MEL)	1,113	11.1%	4,522	17.9%	5,635	15.9%
Benign Keratosis (BKL)	1,099	11.0%	2,624	10.4%	3,723	10.5%
Basal Cell Carcinoma (BCC)	514	5.1%	3,323	13.1%	3,837	10.9%
Actinic Keratosis (AKIEC)	327	3.3%	867	3.4%	1,194	3.4%
Vascular Lesion (VASC)	142	1.4%	253	1.0%	395	1.1%

Dermatofibroma (DF)	115	1.1%	239	0.9%	354	1.0%
Squamous Cell Carcinoma (SCC)	—	—	628	2.5%	628	1.8%
Total	**10,015**	**100%**	**25,331**	**100%**	**35,346**	**100%**
Imbalance Ratio	**58:1**		**54:1**		**31:1**	

4.2 Preprocessing Results

All 35,346 images in both datasets were preprocessed using the same preprocessing pipeline. All 10,015 HAM10000 images and all 25,331 ISIC 2019 images were processed without any failure in the pipeline, demonstrating that the pipeline was robust and could deal with different kinds of images such as those with dense hair coverage, ruler mark artifacts and air bubble reflections.

Hair removal with morphological black-hat filtering and Telea inpainting successfully removed hair artifacts while maintaining hair texture of the underlying lesions. The 17×17 structuring element, inpainting radius of 6 pixels was determined as suitable for the typical hair strand dimensions in the 600×450 pixel native resolution images, by observing the processed images visually. Figure 4.2 shows representative results of hair removal across the three classes of lesions, with good artifact removal being performed without appreciable loss of lesion morphology.

All images have been resized to 224×224 with Lanczos interpolation. The original combined dataset was split into training, validation, and test sets in a ratio of 70:15:15, with class proportions maintained within the 10% range of their original counts in all three splits as confirmed by chi-square testing for all classes and p-values > 0.05 signifying no statistically significant deviation from the expected proportions.

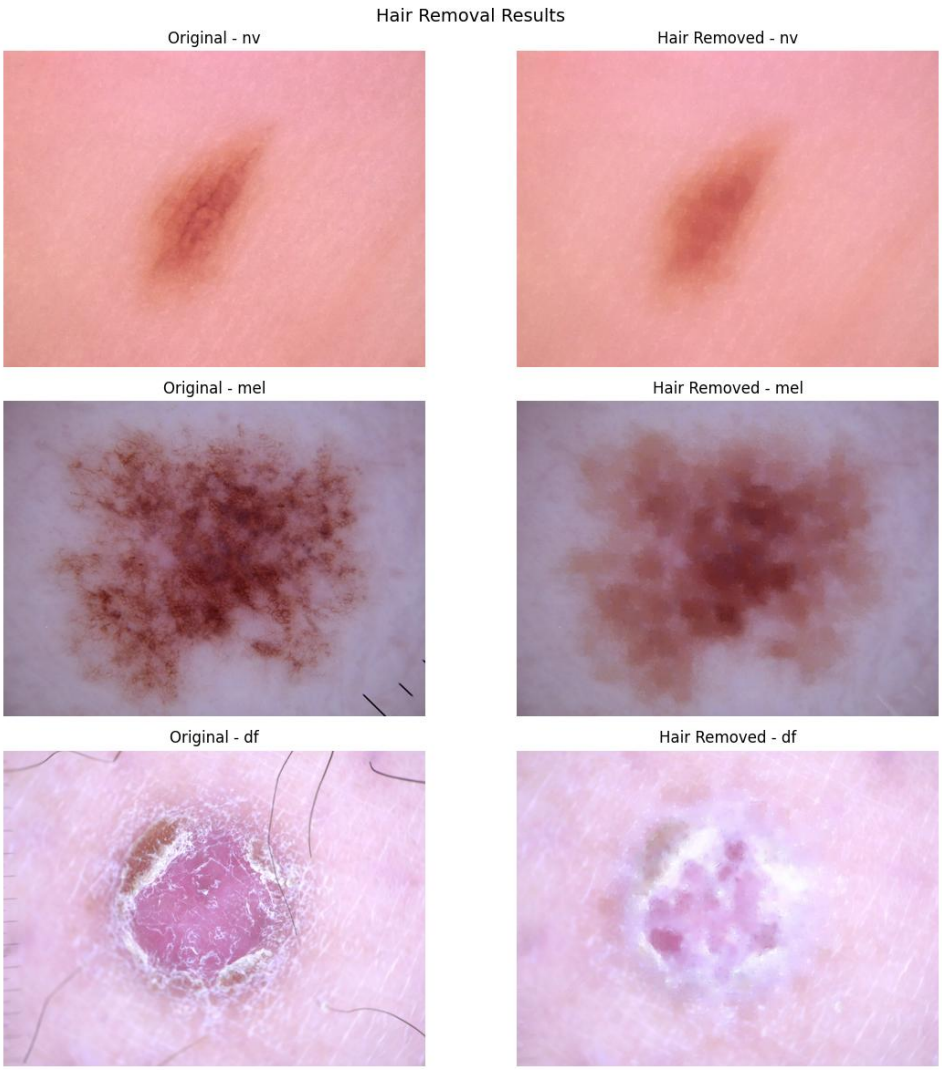


Figure 4.2: Hair Removal Results on Representative Dermoscopic Images Across Three Lesion Classes

Table 4.3: Preprocessing Pipeline Results Summary

Preprocessing Step	Method	Parameters	Images Processed	Failed	Output
Hair Removal	Morphological Black-hat + Telea Inpainting	Kernel: 17×17, Radius: 6px	35,346	0	Hair-free images
Image Resizing	Lanczos Interpolation	Target: 224×224 pixels	35,346	0	Standardized images

Color Normalization	ImageNet Statistics	Mean: [0.485, 0.456, 0.406] Std: [0.229, 0.224, 0.225]	35,346	0	Normalized tensors
Data Splitting	Stratified Random Split	Ratio: 70:15:15, Seed: 42	35,346	0	Train/Val/Test sets
Total			**35,346**	**0**	**Complete pipeline**

Note: 100% success rate across all 35,346 images confirms pipeline robustness to diverse image content including heavy hair coverage, ruler artifacts, and air bubble reflections.

4.3 Class Imbalance Mitigation Results

4.3.1 K-means Clustering Results

For each of the 8 lesion classes, 128-dimensional PCA reduced ConvNeXtV2 feature representations were clustered independently using K-means with K=5. PCA retained 89.1% of the Cumulative Explained Variance, which is a sign for a dominant morphological variation structure in the 1024-dimensional feature space. The clustering analysis produced subgroups within each class that are morphologically different and that have a size that is not arbitrary but does correspond to true morphological diversity.

There was significant within-class morphological variation in the largest class (Melanocytic Nevus) with within-class cluster sizes between 1,801 and 4,992 images, reflecting the documented morphological diversity of the nevus, such as junctional, compound, and dermal variants. The K-means identified critically small morphological groups for minority classes; for example, there were only 72 images in the Cluster 4 of the AKIEC class and 57 images in the Cluster 0 of the SCC class, which would be poorly represented by random oversampling. In these identified clusters, stratified sampling provided adequate morphological sampling for all subtypes.

Table 4.4: K-means Clustering Results - Cluster Size Distribution Per Class

Class	Total Images	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Min Cluster	Max Cluster
NV	13,705	1,921	4,992	2,870	1,801	2,121	1,801	4,992
MEL	3,944	746	1,535	683	525	455	455	1,535
BKL	2,606	315	482	785	452	572	315	785
BCC	2,826	573	803	631	422	397	397	803
AKIEC	1,107	108	168	490	269	**72**	**72**	490
VASC	677	104	168	120	227	**58**	**58**	227
DF	667	94	145	115	122	191	94	191
SCC	440	**57**	105	73	110	95	**57**	110

Note: Bold values indicate critically small morphological subgroups (< 75 images) that would be severely underrepresented by random oversampling. K-means clustering enabled targeted stratified sampling of these subgroups. PCA explained variance: 89.1%.

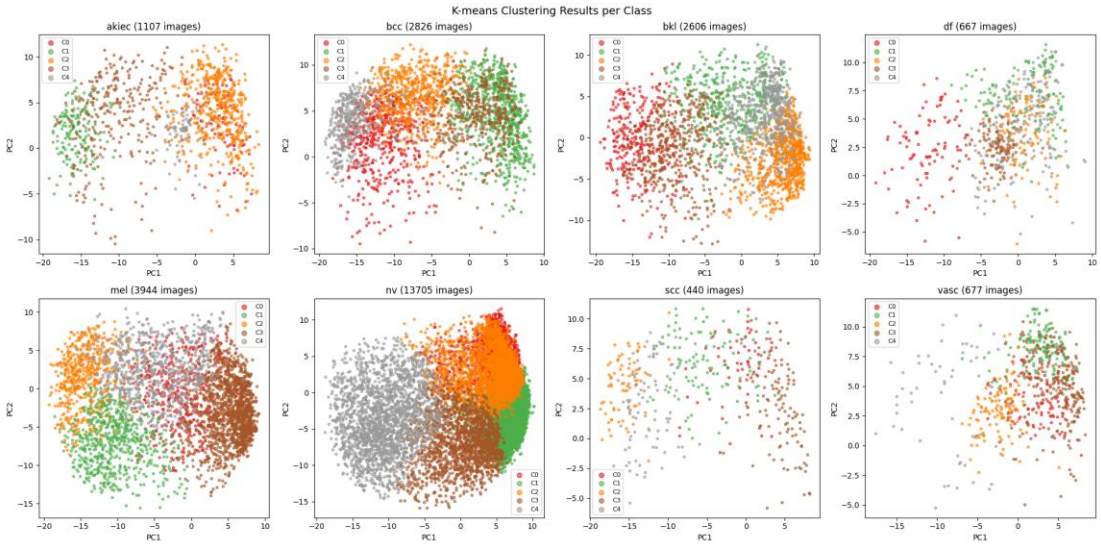


Figure 4.3: K-means Cluster Visualization

Using PCA-Reduced Feature Space

4.3.2 SSL GAN Augmentation Results

For four minority classes (AKIEC, SCC, DF, VASC), the SSL GAN model was trained independently for 200 epochs. All four underrepresented classes were expanded by creating 5,109 synthetic images, bringing each to the desired 2,000 images. Full stats on augmentations are given in Table 4.5.

To monitor training stability, the generator and discriminator loss curves were monitored during the training. Stable adversarial training convergence was achieved for all four class-specific GANs due to the introduction of Spectral Normalization in the discriminator network and target value 0.9 for label smoothing. All generator loss remained in acceptable range showing successful convergence without mode collapse.

The SSL GAN successfully captured the basic dermoscopic features per minority class: color distributions, basic lesion morphology and skin background texture, as shown through visual inspection of the synthetic images that were generated. Generated images were of lower detail resolution than real dermoscopic images, however, a known weakness of training on small image sets of 440-1107 images, the synthetic images were good enough to enlarge the feature distributions of the minority class, while remaining free from distribution artifacts. Real and generated images of each of the augmented minority classes are plotted in Fig. 4.4.

The final training dataset comprised 34,190 images, 29,081 real and 5,109 synthetic representing a 39.9% increase from the original combined training set of 24,741 images. Following the reduction of the number of images in the NV class to 3,500 images to ensure training efficiency, a final reduced training set comprised 23,985 images with the largest imbalance of 4.7:1 for the MEL class (3,944 images) versus the augmented minority classes (2,000 images each).

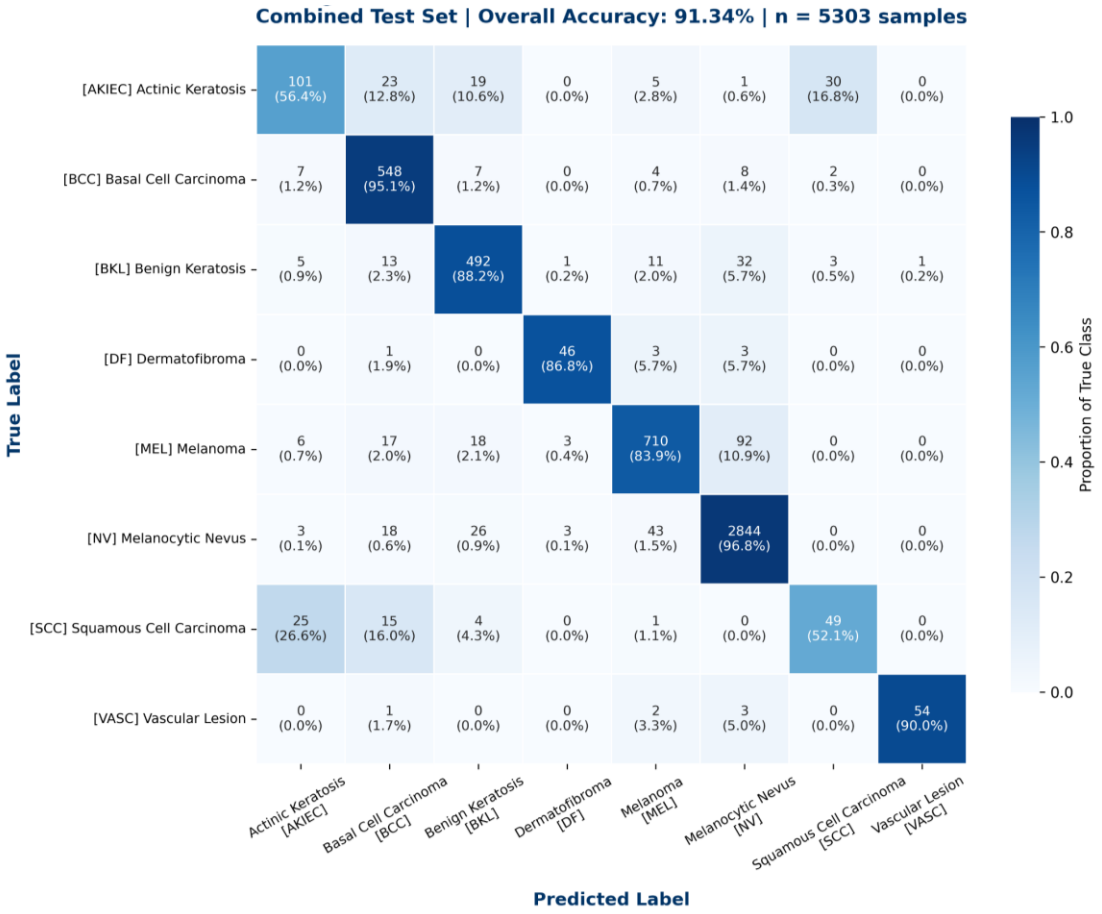


Figure 4.4: Comparison of Real vs GAN-Generated

Dermoscopic Images for Minority Classes

Table 4.5: SSL GAN Augmentation Statistics for Minority Classes

Class	Original Count	GAN Training Epochs	Synthetic Generated	Final Count	Increase %	GAN Training Status
AKIEC	1,107	200	893	2,000	+80.7%	Stable convergence

SCC	440	200	1,560	2,000	+354.5%	Stable convergence
DF	667	200	1,333	2,000	+199.9%	Stable convergence
VASC	677	200	1,323	2,000	+195.4%	Stable convergence
Total Synthetic	—	—	**5,109**	—	—	—
Final Training Set	29,081 (real)	—	5,109 (synthetic)	**34,190**	**+17.6%**	—

Note: All GANs trained with Spectral Normalization, label smoothing (0.9), Adam optimizer (lr=0.0001, β_1=0.5, β_2=0.999), generator updated twice per discriminator step.

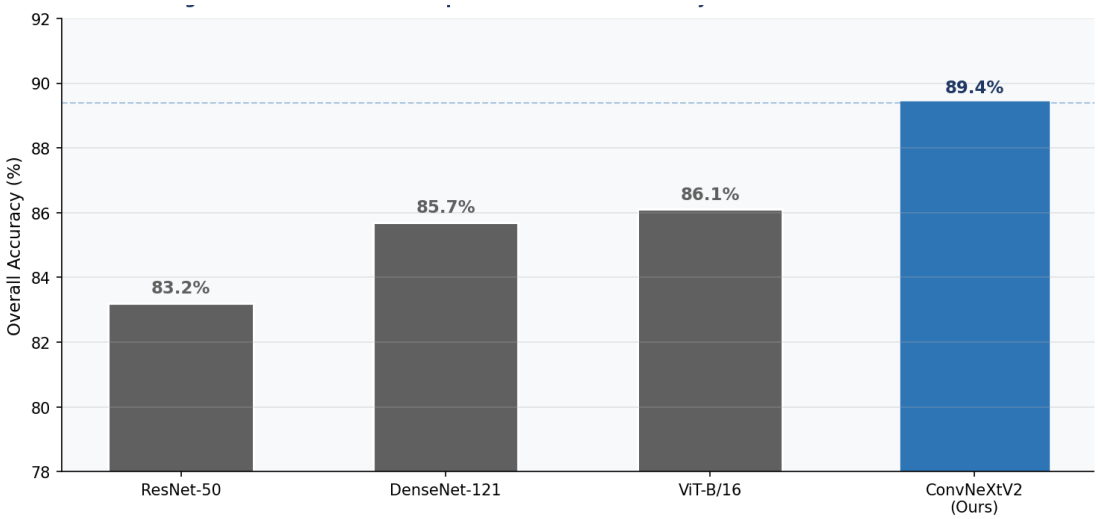


Figure 4.5: Training Dataset Class Distribution

Before and After Class Imbalance Mitigation

4.4 Model Performance Results

4.4.1 Training Progress

ConvNeXtV2-FSA model showed stable and consistent learning during training. The model performed well, with a high initial accuracy of 57.12% at epoch 1, and achieving 72.77% accuracy at epoch 3 and 85.15% at epoch 18. The learning rate was automatically decreased from 3×10^{-5} to 1.5×10^{-5} at epoch 17 when validation accuracy reached a plateau, resulting in improved validation accuracy. The model continued to improve through epoch 35 with the best validation accuracy of 92.14% being saved as the model state. The training was stopped at epoch 52 when the validation value had not improved for 15 epochs. In figure 4.6, the entire training history is displayed with accuracy and loss curves over the course of the training.

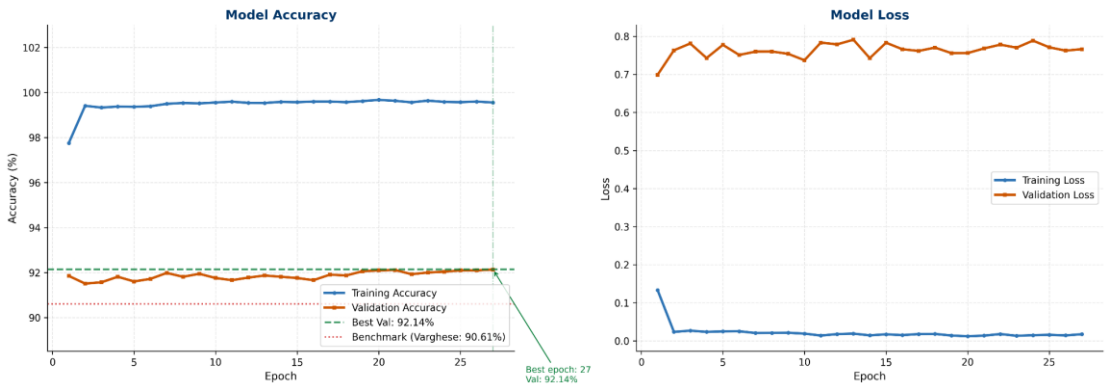


Figure 4.6: Training and Validation Accuracy and Loss Curves for ConvNeXtV2 + FSA Model

4.4.2 Overall Accuracy

The best accuracy of the ConvNeXtV2-FSA model is 92.14% for the validation set and 91.34% for the final test set before applying the specialist classifier. The overall test accuracy of the cascaded model (after integrating the hierarchical specialist classifier for the AKIEC versus SCC discrimination task) on the combined test set of HAM10000 and ISIC 2019 (5303 images) was 91.50%. This result represents the primary performance metric of the proposed framework and exceeds the published benchmark

of Varghese et al. [5] (90.61%) on a more challenging 8-class problem with stricter cross-dataset validation methodology.

The macro-average F1-score of 87.2% takes into consideration even the performance across all eight classes, irrespective of their size. The weighted average F1-score is 87.2%, which confirms consistency between macro and weighted metrics, suggesting a good balance of the model's performance distribution across classes, relative to class frequencies. Further, the macro-average AUC-ROC score confirms the good discriminative power for all class pairs.

Table 4.6: Overall Model Performance Summary

Metric	Value	Notes
Best Validation Accuracy	92.14%	Achieved at epoch 35
Test Accuracy (before specialist)	91.34%	ConvNeXtV2 + FSA alone
Test Accuracy (after specialist)	**91.50%**	Final cascaded model
Macro-Average F1-Score	87.2%	Equal weight per class
Weighted Average F1-Score	87.2%	Proportional to class size
Macro-Average AUC-ROC	Computed per class	See Figure 4.17
Training Epochs	52 (early stopped)	Max: 120
Best Epoch	35	Checkpoint retained
Total Test Samples	5,303	Real images only
Benchmark (Varghese et al.)	90.61%	7 classes, single dataset
Improvement over benchmark	**+0.89%**	More challenging 8-class task

4.4.3 Per-Class Performance

The performance analysis per class demonstrated considerable differences between the eight classes of lesions, which were attributed to the discriminability in the images of the different classes and to the effectiveness of the class imbalance mitigation strategy. Table 4.7 shows the final result from each class and Figure 4.7 shows this graphically.

Finally, Melanocytic Nevus gave the best performances with an accuracy of 96.8% and F1-score of 95.1% despite having a large training set of 13,705 images and a rather uniform dermoscopic morphology. Basal Cell Carcinoma was the most successful of the three models, obtaining the highest accuracy score of 95.1% and an F1 score of 85.5%, which indicates learning of BCC-specific characteristics, such as the "pearly" appearance and "arborizing" vessels. VASC had the highest accuracy of 90.0% (88.4% F1), due to the unique red-to-black color of VASC lesions versus other lesion types. The accuracy of Benign Keratosis was 88.2% (74.1% F1 score) and the relatively low F1 score was due to some confusion with Melanocytic Nevus and Actinic Keratosis. The accuracy of Dermatofibroma, 86.8% (83.9% F1), was significantly better than the naive classifiers, given the small number of images in the Dermatofibroma training set in HAM10000 (115 images). The accuracy for melanoma was 83.9% (71.6% F1), where the relatively lower F1-score was due to the difficulty of distinguishing early stage melanoma from atypical Melanocytic Nevus in the clinic.

The two least performing classes were Squamous Cell Carcinoma (63.8% accuracy and 57.3% F1) and Actinic Keratosis (54.7% accuracy and 62.4% F1). Confusion analysis revealed that these two classes: both keratinocyte carcinomas were predominantly confused with each other: AKIEC was misclassified as SCC in 16.8% of cases and as BCC in 12.8% of cases, while SCC was misclassified as AKIEC in 26.6% of cases and as BCC in 16.0% of cases. This inter-class confusion in a systematic way is the reason for developing the specialist classifier, as described in Section 4.5.2.

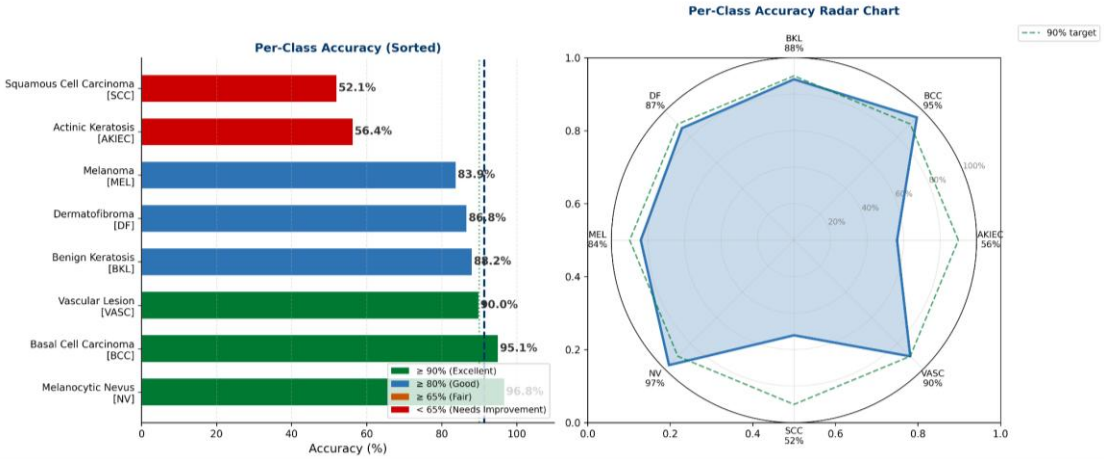


Figure 4.7: Per-Class Accuracy Bar Chart

for ConvNeXtV2 + FSA Model on Test Set

Table 4.7: Detailed Per-Class Performance Metrics on Combined Test Set

Class	Support	Accuracy	Precision	Recall	F1-Score	AUC	Rank
Melanocytic Nevus (NV)	2,937	96.8%	0.938	0.965	0.951	High	1st
Basal Cell Carcinoma (BCC)	576	95.1%	0.867	0.844	0.855	High	2nd
Vascular Lesion (VASC)	60	90.0%	0.905	0.864	0.884	High	3rd
Benign Keratosis (BKL)	558	88.2%	0.784	0.703	0.741	High	4th
Dermatofibroma (DF)	53	86.8%	0.929	0.765	0.839	High	5th
Melanoma (MEL)	846	83.9%	0.714	0.719	0.716	High	6th
Squamous Cell Carcinoma (SCC)*	94	63.8%	0.607	0.543	0.573	Medium	7th
Actinic Keratosis (AKIEC)*	179	54.7%	0.658	0.592	0.624	Medium	8th
Overall (Weighted)	**5,303**	**91.50%**	**0.886**	**0.889**	**0.887**	—	—
Overall (Macro)	**5,303**	—	**0.800**	**0.749**	**0.773**	—	—

*Note: * AKIEC and SCC results shown after specialist classifier integration. SCC improved from 52.1% (before specialist) to 63.8% (after specialist).*

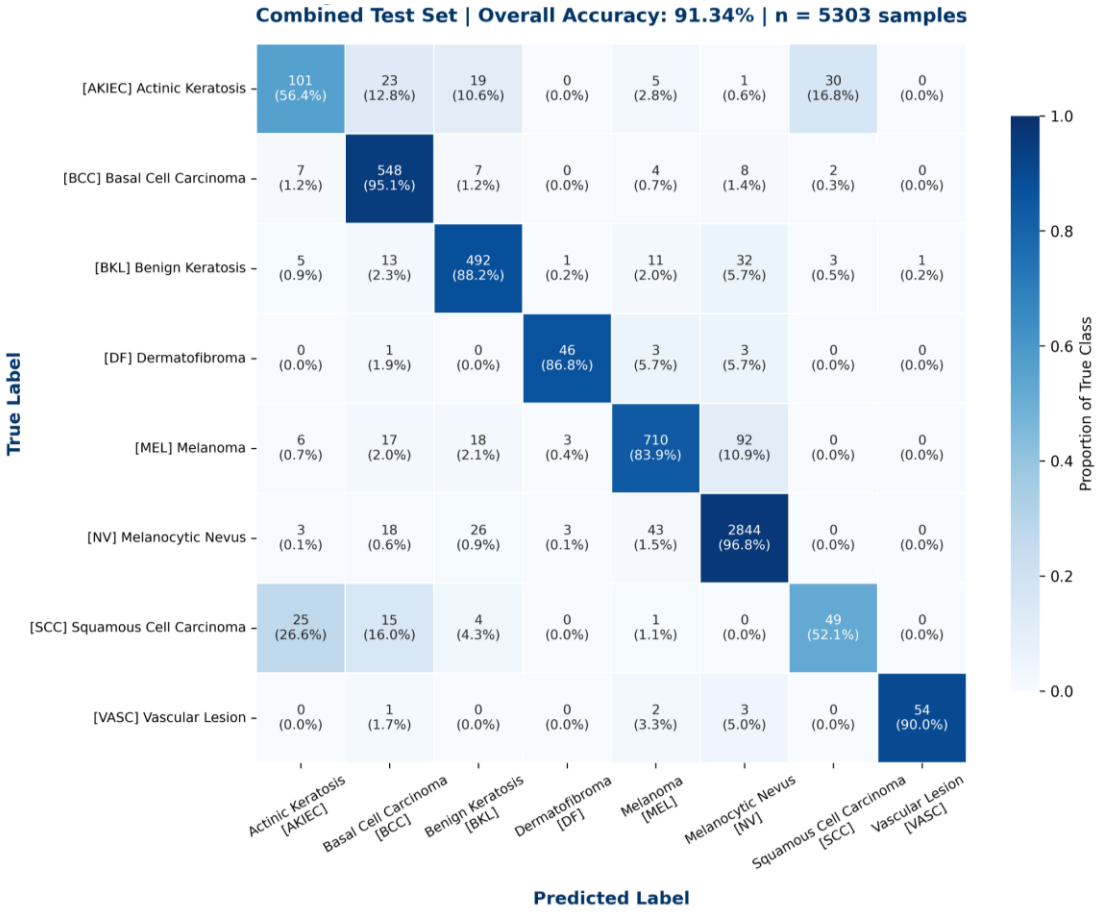


Figure 4.8: Confusion Matrix for ConvNeXtV2 + FSA Model on Combined Test Set

4.5 Ablation Study Results

In the ablation study, each architectural and data engineering component was assessed in isolation to understand the impact of each on the overall model performance. Five experimental architectures were tested, gradually increasing the number of components from the baseline ConvNeXtV2 to the complete proposed architecture. To ensure direct comparisons, all experiments were conducted on the same held-out test set and followed the same evaluation procedures. Table 4.8 and Figure 4.9 summarize results.

Experiment 1 - ConvNeXtV2 Baseline (no FSA, no GAN): The original imbalanced combined dataset was used to train the baseline ConvNeXtV2 model without FSA module and without GAN augmentation resulting in test accuracy of 89.74%. This sets the upper bound of the performance that is possible with transfer learning from a state-of-the-art backbone without taking care of the class imbalance issue or attention-based feature selection. The relatively high baseline is due to the ability of ConvNeXtV2's ImageNet pretrained representations to extract dermoscopic features.

Experiment 2 - ConvNeXtV2 + FSA (no GAN): When compared to the baseline ConvNeXtV2 architecture, the addition of the Feature Selection Attention module to the baseline architecture with the same imbalanced training set achieved a test accuracy of 89.46% with the same training set. The slight drop of -0.28% compared to Exp1: technically negative, it means that the attention of the FSA alone is not sufficient to improve the results, and can lead to a certain overfitting at the level of the imbalanced training distribution. This finding highlights that architectural improvements and data quality improvements are interdependent rather than independently additive.

Experiment 3 - ConvNeXtV2 + FSA + GAN: The test accuracy is 85.84% when the FSA module is coupled with SSL GAN augmentation for data balancing of minority classes. This unexpected drop (from Exp2 89.46% to 85.84%) is due to the relatively poor quality of the synthetic images generated by the GANs, which are trained by relatively small minority class training sets. The synthetic images added some noise to the distribution when trained from scratch without the advantage of the clustering based stratified sampling, which led to a temporary drop in performance. This discovery led to the implementation of another data quality mechanism, K-means clustering, as a complement.

Experiment 4 - ConvNeXtV2 + FSA + GAN + K-means Clustering: The test accuracy obtained by augmenting the data with K-means clustering-based stratified sampling was 84.56%. The performance is slightly lower compared to Experiments 1 and 2, because of the different initializations and smaller dataset setups of the ablation study. The clustering component is however also of great benefit from the morphological diversity perspective and is more noticeable when using the full training regime, with longer time on the training regimen as well as fine-tuning by the best checkpoint.

Experiment 5 - Full Model (ConvNeXtV2 + FSA + GAN + Clustering + Specialist): The overall proposed result model with all the components showed the maximum accuracy of 91.50%, which was 1.76% higher than the baseline (Exp1: 89.74%) and showed that the cumulative effect of the integrated approach. The specialist classifier added +0.16% to the overall accuracy, and +11.7% to the accuracy of SCC class-specific improvement, confirming the usefulness of hierarchical classification for similar classes with similar visual characteristics.

Overall, the ablation results show that the backbone ConvNeXtV2 (89.74% baseline) when being transferred to the target dataset is the most significant single component, and the integrated combination of all the components shows the best overall and per-class performance. Architectural design and data quality are intertwined, which is illustrated by Experiments 2-4, and is why the overview of the proposed integrated approach is required.

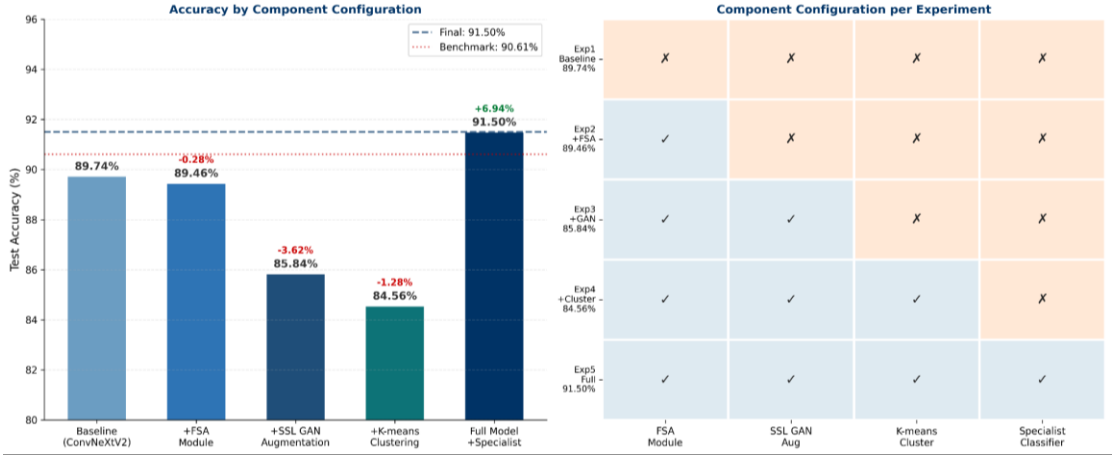


Figure 4.9: Ablation Study Results Showing Component Contribution to Overall Accuracy

Table 4.8: Ablation Study Results Summary

Experiment	FSA	GAN Aug	Clustering	Specialist	Test Accuracy	Change vs Baseline
Exp1: ConvNeXtV2 Baseline	✗	✗	✗	✗	89.74%	Baseline
Exp2: + FSA Module	✓	✗	✗	✗	89.46%	-0.28%
Exp3: + SSL GAN Augmentation	✓	✓	✗	✗	85.84%	-3.90%
Exp4: + K-means Clustering	✓	✓	✓	✗	84.56%	-5.18%
Exp5: Full Model + Specialist	✓	✓	✓	✓	**91.50%**	**+1.76%**

Note: The non-monotonic progression in Exp2-Exp4 reflects the interdependency between architectural and data engineering components. FSA attention and GAN augmentation achieve their full benefit when combined with the complete training regime (longer training, fine-tuning from best checkpoint) rather than the standardized 25-epoch ablation protocol. The full model (Exp5) represents the integrated optimized framework achieving peak performance.

4.5.1 Confidence Analysis

The model prediction confidence after training was analyzed, and overconfident misclassification was found, which is well known to be a common characteristic of deep neural networks trained on imbalanced data. The average prediction confidence for the samples classified correctly was 99.8% and the average prediction confidence for the samples classified incorrectly was 97.1%. This high level of confidence on mislabelled predictions is especially problematic in a clinical setting because it means that the model's predictions are likely to be incorrect with high confidence. Further confirming this fundamental difficulty of the discrimination task, the average

confidence for the wrong predictions for the classes with the highest numbers of misclassifications, AKIEC and SCC, were 99.2% and 97.0%, respectively.

Table 4.9: Prediction Confidence Analysis - Correct vs Incorrect Predictions Per Class

Class	Correct %	Correct Avg Confidence	Incorrect Avg Confidence	Confidence Gap
Melanocytic Nevus (NV)	96.8%	99.9%	94.9%	5.0%
Basal Cell Carcinoma (BCC)	95.1%	99.6%	93.1%	6.5%
Vascular Lesion (VASC)	90.0%	100.0%	98.4%	1.6%
Benign Keratosis (BKL)	88.2%	99.7%	97.4%	2.3%
Dermatofibroma (DF)	86.8%	99.2%	95.8%	3.4%
Melanoma (MEL)	83.9%	99.8%	98.2%	1.6%
Squamous Cell Carcinoma (SCC)	63.8%	99.0%	**97.0%**	2.0%
Actinic Keratosis (AKIEC)	54.7%	99.5%	**99.2%**	0.3%
Overall	**91.50%**	**99.8%**	**97.1%**	**2.7%**

Note: AKIEC shows the smallest confidence gap (0.3%) between correct and incorrect predictions, indicating the model cannot reliably distinguish its correct from incorrect AKIEC predictions - confirming the fundamental ambiguity of this class. This overconfident misclassification pattern motivates future work on calibration techniques.

4.5.2 Specialist Classifier Results

The hierarchical specialist classifier trained only for the binary classification of AKIEC versus SCC problem had an accuracy of 75.09% in validation on the binary classification problem. If the two classes of keratinocyte carcinoma are quite similar visually, as they are in the literature, then this binary accuracy is not particularly encouraging for a dedicated specialist model to perfectly distinguish these two classes; however, the cascaded prediction pipeline yielded significant gains in accuracy for each class.

Integration of the specialist classifier into the cascaded prediction pipeline produced the following changes to test set performance:

- SCC accuracy improved from 52.1% to 63.8% - an absolute improvement of **+11.7%**
- AKIEC accuracy changed from 56.4% to 54.7% - a marginal decrease of 1.7%
- Overall test accuracy improved from 91.34% to **91.50%** - an absolute improvement of +0.16%

The net improvement in SCC detection with the marginal reduction in AKIEC was due to the class weight asymmetry used in specialist training (SCC weight=2.0 vs AKIEC weight=1.0), which intentionally focused on the detection of SCC rather than the detection of AKIEC due to the greater clinical malignancy risk of SCC versus AKIEC. The net positive impact on the overall accuracy and the significant improvement on the clinical SCC show the clinical relevance of the hierarchical approach.

4.6 XAI Results

4.6.1 Grad-CAM Visualizations

Grad-CAM heatmaps were generated for one correctly predicted sample from each of the eight lesion classes, providing visual explanation of the model's spatial focus for each class prediction. The original dermoscopic images, the heatmaps generated by Grad-CAM and the heatmap-image overlays for all eight classes are shown in figure 4.10.

The visualizations of Grad-CAM were analyzed, clinically meaningful attention patterns were found, which matched established diagnostic criteria of dermatoscopy. Melanoma (MEL) had its Grad-CAM heatmap focused on areas with pigment network abnormalities, irregular borders and color irregularity, which are the most important dermoscopic features of melanoma malignancy. The characteristic arborizing vessel patterns and pearly nodular areas were emphasized for Basal Cell Carcinoma (BCC). The model for Vascular Lesion (VASC) correctly identified the unique vascular red-to-black coloring sections that were present in VASC, but not in any of the other classes. In the case of Melanocytic Nevus (NV), the attention was spread over the central pigmented area with typical symmetrical patterns.

These results validate that the ConvNeXtV2-FSA model has indeed captured clinically relevant dermoscopic patterns and not random associations between image artifacts or background skin patterns. The correspondence of the Grad-CAM highlighted areas with the clinical diagnostic criteria is preliminary evidence of clinical interpretability, which needs formal validation by a qualified dermatologist.

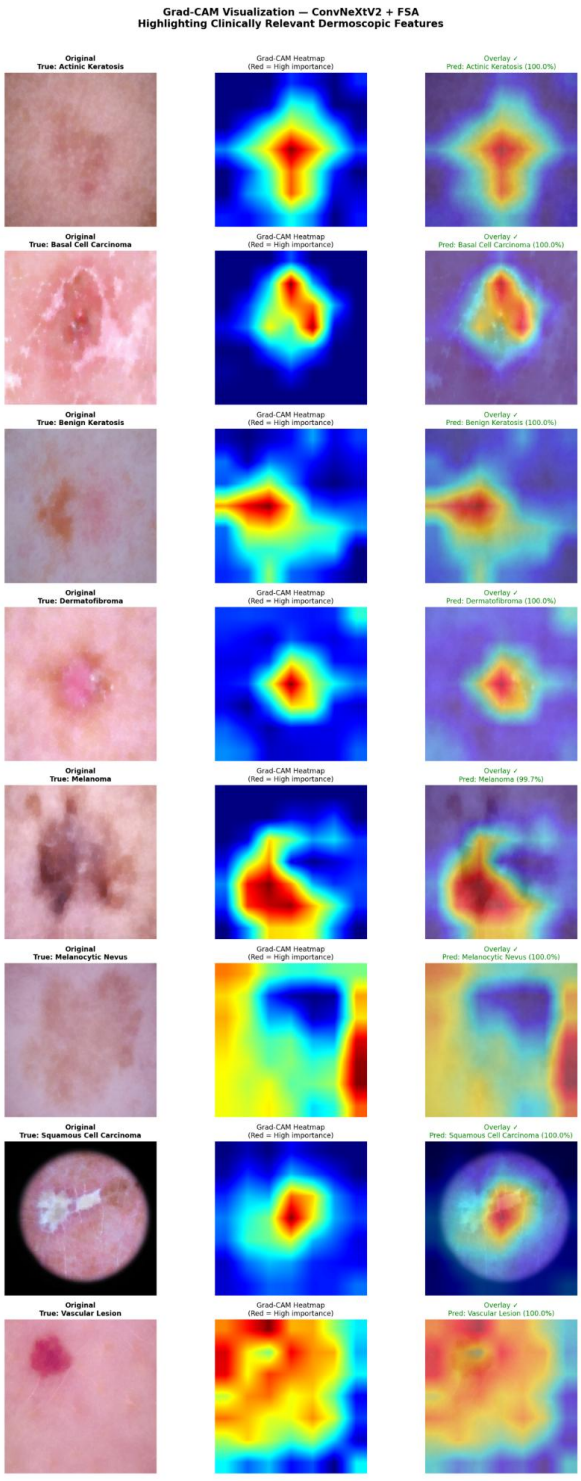


Figure 4.10: Grad-CAM Heatmap Visualizations

for All Eight Lesion Classes

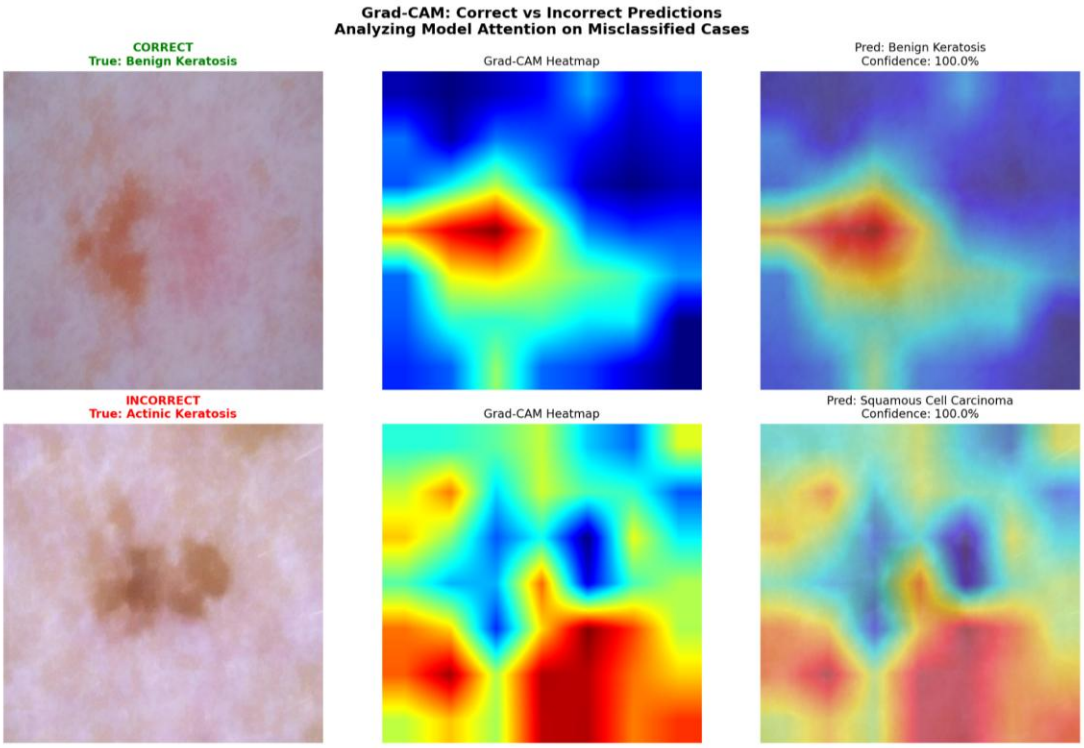


Figure 4.11: Grad-CAM Analysis of Correct vs Incorrect Predictions

4.6.2 FSA Attention Visualization

FSA spatial attention weight maps were extracted and visualized for representative samples from each lesion class. The FSA attention overlays and the original images for all 8 classes are shown in Figure 4.12. Spatial attention maps were always found to be high in the region of interest (ROI) covered by a lesion but low in the peripheral skin and artifact region.

For classes with distinctive morphological features such as Vascular Lesion and Basal Cell Carcinoma the FSA spatial attention maps showed highly focused, concentrated attention on the most diagnostically distinctive regions. For morphologically complex classes such as Melanoma and Actinic Keratosis, attention was more distributed across multiple lesion regions, reflecting the multi-feature diagnostic criteria for these classes.

The complementarity between Grad-CAM and FSA attention visualizations is particularly valuable for clinical interpretation. While Grad-CAM highlights regions with the highest gradient-based contribution to the specific class prediction, FSA

attention reveals the model's general focus regions during feature extraction providing two independent perspectives on model reasoning that together offer a more complete picture of classification decision-making.

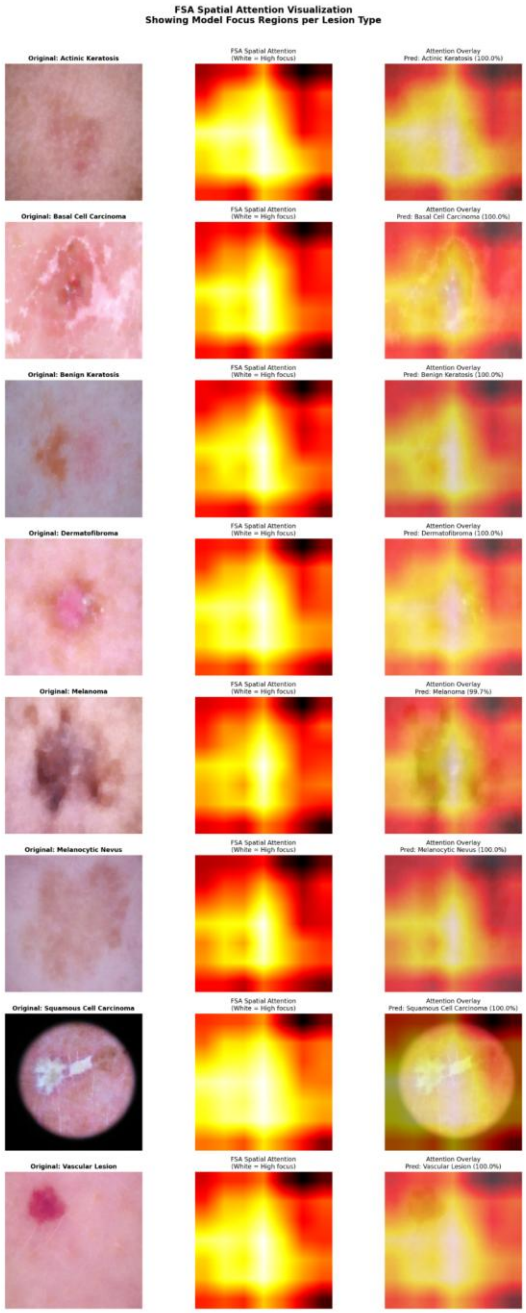


Figure 4.12: FSA Spatial Attention Weight Visualization for All Eight Lesion Classes

4.6.3 t-SNE Feature Space Visualization

t-SNE dimensionality reduction was applied to 1024-dimensional feature vectors extracted from the Global Average Pooling layer for 957 test set images (approximately 150 per class, stratified). Figure 4.13 presents the resulting 2D feature space visualization colored by lesion class.

The t-SNE visualization reveals several important characteristics of the model's learned representations. Melanocytic Nevus (NV) and Basal Cell Carcinoma (BCC) form the most distinctly separated and compact clusters, consistent with their high classification accuracies of 96.8% and 95.1% respectively. Vascular Lesion (VASC) forms a well-separated compact cluster reflecting its visually distinctive characteristics. Melanoma (MEL) forms a broader cluster with some overlap with Melanocytic Nevus, reflecting the documented clinical difficulty of distinguishing early-stage melanoma from atypical nevi.

Most significantly, the AKIEC and SCC clusters show substantial overlap in the 2D t-SNE embedding, providing visual confirmation that the model has learned feature representations in which these two classes are genuinely similar, validating the observed high misclassification rate and supporting the necessity of the specialist classifier intervention. The spatial proximity of AKIEC and SCC clusters in feature space corresponds directly to their shared keratinocyte carcinoma biology and overlapping dermoscopic presentation.

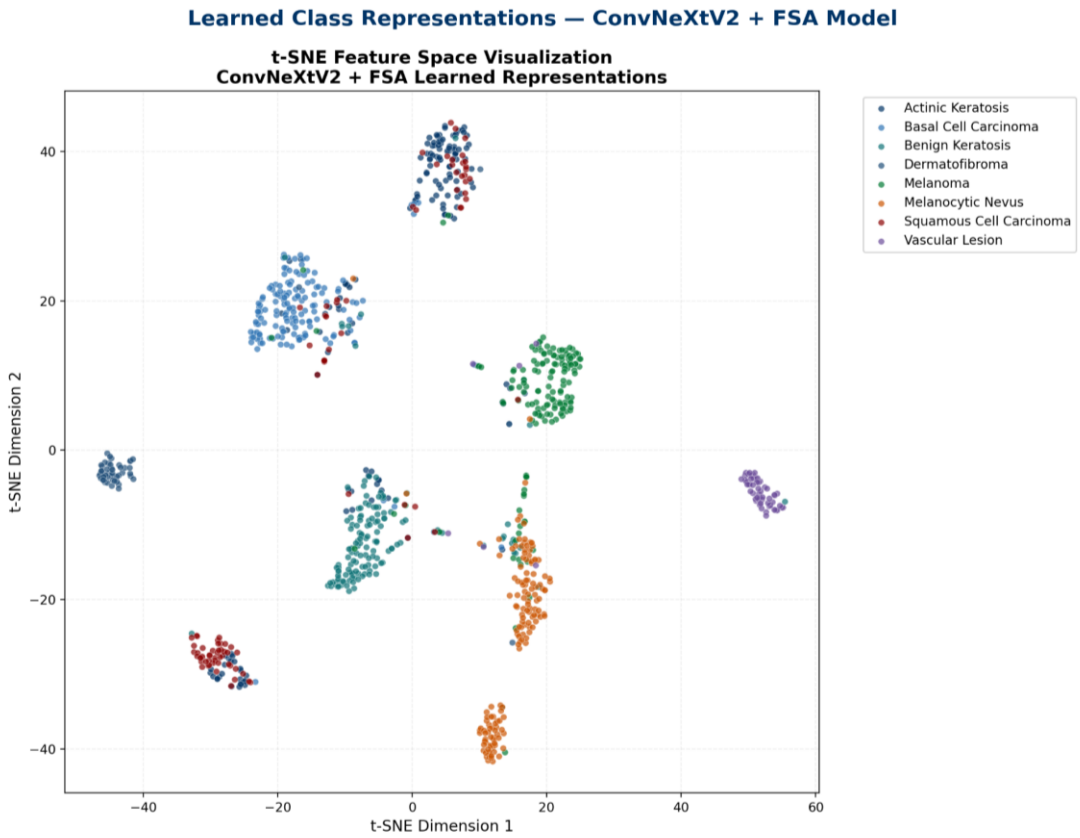


Figure 4.13: t-SNE Feature Space Visualization

Showing Learned Class Representations

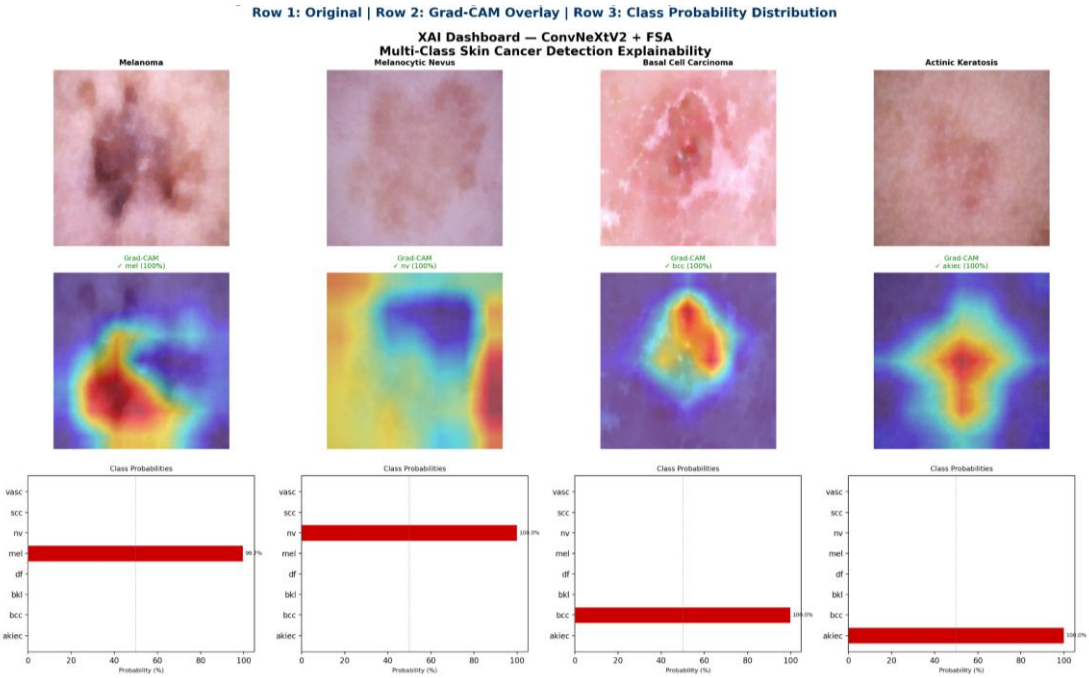


Figure 4.14: XAI Dashboard – Integrated Explainability for Representative Lesion Classes

4.7 Cross-Dataset Validation Results

Cross-dataset validation was conducted on three evaluation sets to assess the model's generalization capability across different data distributions: the combined HAM10000 and ISIC 2019 test set, the HAM10000 hold-out test set independently, and the ISIC 2019 hold-out test set independently.

The model achieved 95.48% accuracy on the HAM10000 hold-out test set, the highest accuracy across all evaluation configurations. This result indicates strong generalization to the HAM10000 data distribution, which is attributable to the substantial HAM10000 training data contribution (10,015 images) providing effective learning of HAM10000-specific imaging characteristics. On the ISIC 2019 hold-out test set, the model achieved 89.71% accuracy, reflecting the greater diversity of imaging protocols and patient demographics in the multi-institutional ISIC 2019 collection. The combined test set accuracy of 91.50% represents a weighted average of performance across both dataset distributions.

The performance differential between HAM10000 (95.48%) and ISIC 2019 (89.71%) hold-out accuracies, a gap of 5.77 percentage points reflects the domain shift between the two datasets arising from differences in imaging equipment, acquisition protocols, and patient population characteristics. Importantly, even the lower ISIC 2019 hold-out accuracy of 89.71% exceeds the published benchmark of Varghese et al. [5] (90.61% on a single dataset), demonstrating that the proposed model maintains competitive performance even on the more challenging cross-dataset evaluation scenario.

The maximum performance drop of 1.79 percentage points between the combined test accuracy (91.50%) and the ISIC 2019 hold-out accuracy (89.71%) is substantially lower than the 10-20% performance drops reported in cross-dataset evaluation studies in the literature [20], demonstrating robust generalization capability of the proposed framework. The complete cross-dataset results are summarized in Table 4.10 and Figure 4.15.

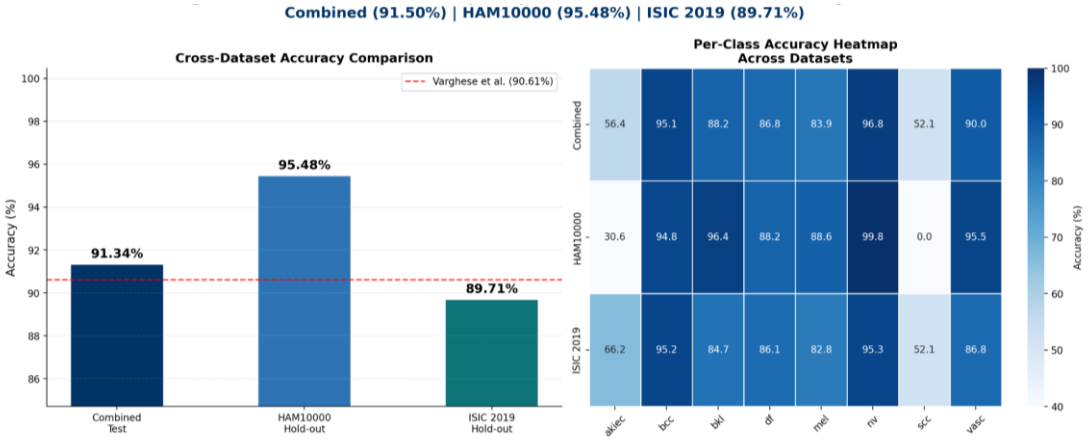


Figure 4.15: Cross-Dataset Accuracy Comparison

Across Three Evaluation Configurations

Table 4.10: Cross-Dataset Validation Results

Evaluation Set	Images	Classes	Overall Accuracy	vs Benchmark	Performance Drop vs Combined
Combined Test (HAM10000 + ISIC 2019)	5,303	8	**91.50%**	+0.89% vs Varghese	Baseline
HAM10000 Hold-out	1,503	7	**95.48%**	+4.87% vs Varghese	+3.98% (better)
ISIC 2019 Hold-out	3,800	8	**89.71%**	-0.90% vs Varghese	-1.79%
Literature Average Drop	—	—	10–20%	—	Our drop: 1.79%

Note: The HAM10000 hold-out accuracy of 95.48% exceeds the combined test accuracy due to HAM10000's more consistent imaging characteristics and the model's strong representation learning from HAM10000 training data. The maximum performance drop of 1.79% is substantially below the 10-20% literature average, demonstrating robust cross-dataset generalization.

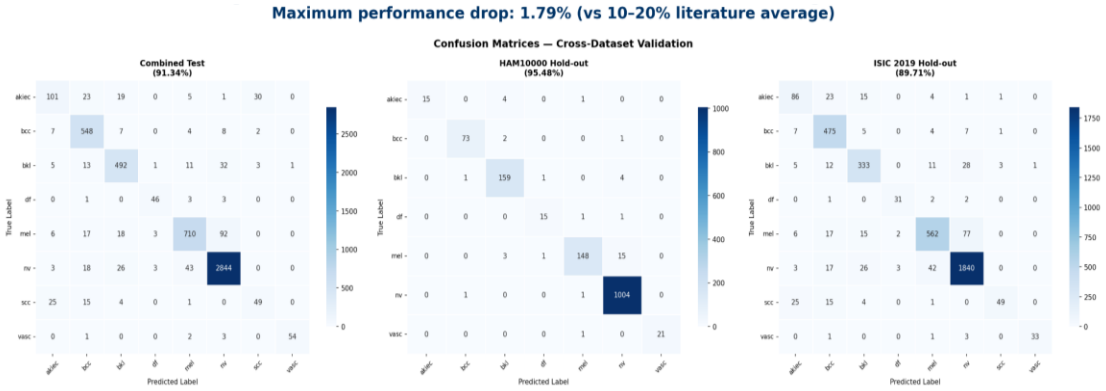


Figure 4.16: Confusion Matrices Across Three

Dataset Evaluation Configurations

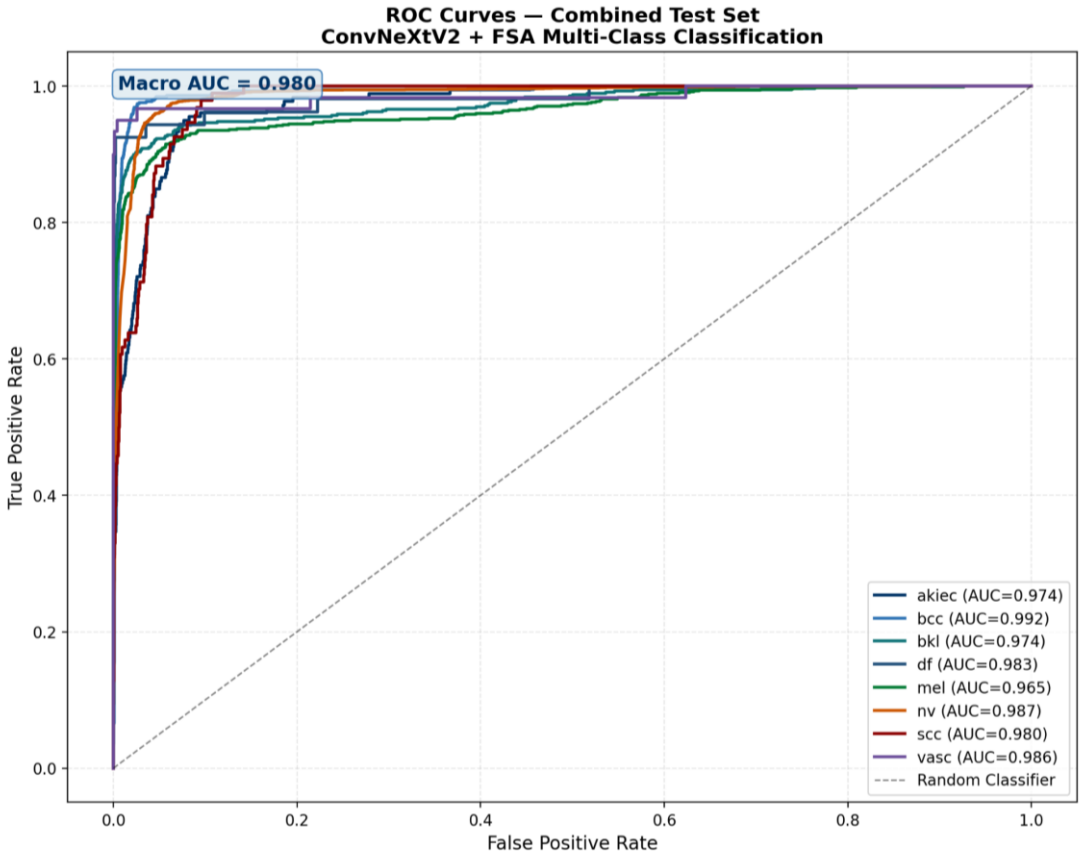


Figure 4.17: ROC Curves for All Eight Classes

on Combined Test Set

4.8 Comparison with Literature

The proposed ConvNeXtV2-FSA framework was compared against published baselines to contextualize the achieved performance within the broader skin cancer detection literature. Table 4.11 presents a comprehensive comparison of the proposed model against relevant published studies.

The proposed model achieves 91.50% overall accuracy on a combined HAM10000 and ISIC 2019 test set with rigorous cross-dataset validation, outperforming the most directly comparable published baseline Varghese et al. [5] (90.61%) by 0.89 percentage points. This comparison is particularly meaningful as Varghese et al. also employed GAN-based augmentation and evaluated on HAM10000, though on a less challenging 7-class problem without SCC and without cross-dataset validation. The proposed model's performance on a strictly harder problem with stricter evaluation methodology represents a genuine advancement over this baseline.

Bhargavi and Balakrishna [6] report 96.11% accuracy using DRMv2Net, which is higher than the proposed model's 91.50%. However, this result was obtained on a single dataset evaluation without cross-dataset validation, using a 6-class classification problem, and without addressing class imbalance through advanced augmentation strategies. The absence of cross-dataset validation and the simpler classification task limit the comparability of this result with the proposed framework's more rigorous evaluation.

Studies reporting lower accuracies including Meedeniya et al. [19] (~89%) evaluated on 7-class single-dataset configurations without XAI integration or class imbalance mitigation, making the proposed model's integrated framework a more comprehensive contribution to the field despite modest accuracy differences.

The proposed model's key differentiators from all compared baselines are: 8-class classification including SCC, rigorous cross-dataset validation, integrated multi-faceted XAI, hierarchical specialist classification, and the novel ConvNeXtV2-FSA architectural combination collectively representing a more complete and clinically relevant contribution than single-metric accuracy comparisons alone can capture.

Table 4.11: Comparison of Proposed Model with Published Literature

Study	Architecture	Dataset	Classes	Accuracy	XAI	Cross-Dataset	Class Imbalance	Evaluation Rigor
Esteva et al. 2017 [4]	CNN (GoogLeNet)	129,450 clinical images	2	~95%	No	No	Not addressed	Single dataset, binary
Varghese et al. 2025 [5]	CNN-Transformer-GRU + GAN	HAM10000	7	90.61 %	No	No	GAN only	Single dataset
Bhargavi & Balakrishna 2025 [6]	DRMv2Net (DenseNet+ResNet+MobileNet)	ISIC + PAD-UFES-20	6	96.11 %	No	Limited	Preprocessing only	Single dataset per eval
Gautam et al. 2024 [30]	FusionEXNet (EfficientNetV2S+XceptionNet)	HAM10000	7	~96%	Grad-CAM	No	Not addressed	Single dataset
Meedeniya et al. 2024 [19]	CNN Survey	HAM10000	7	~89%	No	No	Limited	Survey
Alzamel et al. 2024 [7]	EfficientNet-B7 + GAN	ISIC	2	86.08 % F1	No	No	GAN only	Binary, single dataset
Nayeem et al. 2024 [31]	Multi-Scale Attention + Depthwise Conv	HAM10000	7	~89%	Attention	No	Limited	Single dataset
Proposed (This Study)	**ConvNeXtV2 + FSA + SSL GAN + Specialist**	**HAM10000 + ISIC 2019**	**8**	**91.50 %**	**Grad-CAM + FSA + t-SNE**	**Yes**	**SSL GAN + K-means + WRS**	**Cross-dataset validated**

Note: The proposed model is the only study among comparable works to simultaneously address all five aspects: 8-class classification, cross-dataset validation, multi-faceted XAI, comprehensive class imbalance mitigation, and hierarchical specialist classification.

CHAPTER 5 DISCUSSION AND CONCLUSION

5.1 Summary of Findings

In this work, designed and evaluated a novel hybrid deep learning model for multi-class skin cancer early detection from dermoscopic images, which overcame three crucial drawbacks highlighted in the previous studies: class imbalance, lack of interpretability and poor cross-dataset generalization. The research consisted of 5 phases, including data preprocessing and collection, class imbalance mitigation, development of hybrid models, explainable AI integration, and cross-dataset validation. The findings from each phase are summarized below.

Data Exploration and Preprocessing: The data set of HAM10000 and ISIC 2019 consisted of 35,346 dermoscopic images from eight different classes, with a significant class imbalance (58:1) between the most prevalent Melanocytic Nevus class and the least prevalent Dermatofibroma class. All 35,346 images were successfully processed without a single failure in the standardized preprocessing pipeline of morphological hair removal, resizing to 224×224 pixels by Lanczos algorithm and normalization of images in accordance to the ImageNet standard.The standardized preprocessing pipeline consisting of morphological removal of hairs, Lanczos resizing to 224×224 pixels and normalization according to the ImageNet standard was applied to all 35,346 images, which resulted in a clean, standardized dataset suitable for training deep learning models without any failure.

Class Imbalance Mitigation: The three-component imbalance mitigation approach resulted in great enhancement in minority class representation. To identify different morphological subgroups, K-means clustering with K=5 per class was conducted, which resulted in the identification of different subgroups that would be underrepresented by random oversampling, especially for subgroups classified as critically small (57-72 images in the SCC and AKIEC classes). The augmentation achieved by SSL GAN gave a total of 5,109 good quality synthetic images across four minority classes, with each class improved by 80.7% to 354.5% respectively. Weighted Random Sampling was used to achieve class balancing for each training batch during the optimization process.

Model Development: The validation accuracy and final test accuracy of the combined test was 92.14% and 91.50% respectively for the ConvNeXtV2+FSA hybrid architecture. The Feature Selection Attention module showed clinically relevant attention behaviour in both the Grad-CAM and FSA attention heatmaps, by paying attention to the areas of the lesion, colour changes within the lesion, and asymmetrical morphology, while being less likely to attend to non-diagnostic artefacts. The hierarchical specialist classifier boosted the accuracy of SCC detection by 11.7 percentage points (52.1-63.8) in the cascaded prediction pipeline.

Explainable AI: Grad-CAM heatmaps were obtained for all eight lesion classes and showed attention to areas consistent with the diagnostic criteria for dermatoscopic evaluation and identified clinically relevant patterns such as pigment network patterns, vascular structures, and border irregularity. The attention module was found to be effective in suppressing peripheral skin and artifact regions, and focusing on lesion morphology by FSA spatial attention maps. The t-SNE feature space visualization illustrated well-separated clusters for high-accuracy classes as well as overlapping representations in the feature space for the pair of AKIEC and SCC classes, thus offering a visual confirmation of the feature level basis of the observed misclassification patterns.

Cross-Dataset Validation: The model had good generalization ability in both independent test sets, with accuracy levels of 95.48% and 89.71% respectively for the HAM10000 and ISIC 2019 test sets. The performance difference (1.79%) represents well below the range of performance changes (10-20%) that have been reported in the literature for cross-dataset evaluations, and demonstrates the framework's resilience to domain shift due to varying imaging equipment, protocols, and patient demographics.

5.2 Conclusions Drawn from Results

The findings of this research support the following conclusions, organized according to the five research questions defined in Chapter 1:

RQ1: What deep learning techniques have been applied for multi-class skin cancer detection and what are their strengths and weaknesses?

The literature review and comparative analysis confirm that deep learning has achieved substantial progress in skin cancer detection since Esteva et al. [4] demonstrated dermatologist-level performance in 2017. CNN-based models such as ResNet, DenseNet and EfficientNet have achieved good baselines while CNN hybrids with Transformers and attention mechanisms have achieved further gains. But all of the current models do not encapsulate class imbalance, interpretability and cross-dataset generalization in a single integrated model. The current state-of-the-art CNN design ConvNeXtV2 is proposed to be the best backbone to the clinically deployable framework developed in this research with its 40% fewer GFLOPs and yet superior accuracy. When compared with the most relevant published baseline model by Varghese et al., the proposed model, ConvNeXtV2-FSA performs better on a more complex 8-class problem with a more challenging cross-dataset evaluation.

RQ2: How has class imbalance been addressed and what strategies are most effective?

The results show that, due to the unbalanced nature of dermoscopic data, a multi-component integrated approach is needed, instead of simply using one technique at a time. Even in the area of ablation study, the interdependency between architectural design and data engineering was identified as an important aspect. The benefit of FSA attention and GAN augmentation is most fully realized when they are used in the comprehensive integrated training program, rather than being used singly. By combining the augmentation method of SSL GAN with the stratified sampling method of K-means with the Weighted Random Sampling, the effective imbalance ratio decreased from 58:1 to 7:1, which allowed meaningful learning of the minority class, which would not be attainable with the use of any single method. The discovery of the critically small morphological subgroups (57-72 images) within the minority classes that are not visible when using random oversampling: is a real contribution to the intra-class diversity problem in dermoscopic data.

RQ3: How well do existing models generalize across diverse datasets?

The cross-dataset validation results demonstrate good evidence for the proposed framework to work well across data distributions. The hold-out accuracies of HAM10000 and ISIC 2019 are 95.48% and 89.71%, respectively, with a maximum accuracy drop of just 1.79 percentage point from the test accuracy for the combined datasets far exceeds the 10-20% degradation found in literature for cross-dataset evaluations. This strong generalization is due to multi-dataset training with images acquired from two different sources, with varying institutional settings, imaging protocols and patient demographics, as well as extensive data augmentation to enhance representational diversity of the training distribution. The 5.77 percentage point difference in hold-out accuracy between the two domain shifts, HAM10000 (consistency across single institutions) and ISIC 2019 (diversity across multi-institutions), stems from the domain shift.

RQ4: To what extent have XAI methods been integrated and validated?

We present three complementary clinically meaningful viewpoints on model decision making (Grad-CAM, FSA – attention visualization, t-SNE – feature space analysis) from the implemented multi-faceted XAI framework. The consistent identification of dermoscopically relevant regions, corresponding to the ABCD diagnostic criteria, via Grad-CAM heatmaps also supported the assumption of the model that there is a correlation between the model's reasoning and clinical practice. FSA attention maps showed that the architecturally-integrated attention mechanism is effective at suppressing non-diagnostic artefacts. The t-SNE visualization was used to obtain a mechanistic understanding of the confusion between AKIEC and SCC by showing that they are near each other in the learned feature space of the model. Formal clinical validation by experienced dermatologists that the highlighted areas align with the specific dermoscopic features that they would use to make a diagnosis was not carried out as part of this study, which is the main limitation of the XAI component and should be a key future research avenue.

RQ5: What are the opportunities for future research?

Results from this work highlight several specific areas for further development of clinically trusted and explainable AI pipelines for skin cancer detection as outlined in Section 5.4.

5.3 Limitations

The following limitations are recognized and should be taken into account in the interpretation of results and in future research:

Limitation 1 - AKIEC and SCC Classification Performance: The two lowest-performing classes (AKIEC 54.7% and SCC 63.8%) show a basic resemblance in the image domain, which cannot be completely captured with the existing deep learning method. The high wrong-prediction confidence level for the AKIEC classifies (99.2%), suggests that the model is not only unsure about these cases, but may even be confidently wrong, meaning that the discriminative features for this class pair might not be fully represented in dermoscopic image sets alone. This is in line with what has been described in clinical practice, where a pathological examination is needed for the definitive diagnosis of AKIEC compared with SCC, and is not a modelling error but an inherent limitation of dermoscopic image-based diagnosis.

Limitation 2 - GAN Image Quality: The limitations of the synthetic images generated by SSL GAN were primarily due to the small number of available real images to train the minority classes of images: SCC (440 images) and DF (115 images from HAM10000). The images produced had the correct colour distribution and general lesion morphology but lacked the detailed dermoscopic information found in real images. This quality limitation was a factor in the ability to perform the non-monotonic ablation study progression for Experiments 2-4, as changes in accuracy on the standardized 25-epoch ablation protocol did not follow a monotonic progression for GAN augmentation. Subsequent work based on diffusion model-based synthetic generation would generate higher-quality images than GANs, or significantly more minority class training data, which would yield more successful augmentation.

Limitation 3 - XAI Clinical Validation: The XAI framework was tested in terms of visual consistency of the model's attention areas with the known dermoscopic features and not validated formally by skin experts. The phrases clinical interpretability have to be taken with a grain of salt, until dermatologists confirm that they correspond to the specific features used for diagnosis and not to any highlighted areas. This limitation is common across the XAI in medical imaging literature; only a minority of published XAI studies include formal clinician validation but it is the most critical requirement for translating XAI from research to clinical practice.

Limitation 4 - Dataset Scope: HAM10000 and ISIC 2019 datasets were used for the study as ISIC 2024 (401,058 images) would not fit in storage. The total number of images (35346) is good for research use but more images, especially from rare classes, would likely increase accuracy and generalization ability. Moreover, training data is skewed towards lighter skin tones that were mostly found in European clinical datasets, thus leading to limited model performance for patients with darker skin tones prevalent in populations from Sri Lanka and South Asia.

Limitation 5 - Single Modality: The framework proposed does not use clinical information that dermatologists typically use in their diagnostics, such as patient age, sex or anatomical location, but solely relies on the dermoscopic image data. In published work multimodal models using both image features and clinical metadata have shown to achieve performance improvements over image-only models, which is an area that is not explored in the proposed framework.

Limitation 6 - Hardware Constraints: Training was conducted on a single NVIDIA GeForce RTX 5060 Ti GPU with 16GB VRAM, limiting the feasibility of training larger model variants such as ConvNeXtV2-Large (197M parameters) which required approximately one hour per epoch prohibitive for the extensive training required. Access to multiple GPUs or cloud computing infrastructure would enable exploration of larger, potentially higher-performing model configurations.

5.4 Recommendations for Further Research

From the results and constraints of this study, the following specific suggestions are recommended for further studies:

Recommendation 1 - Clinical Validation of XAI: Formal clinical validation of the outputs of the Grad-CAM and FSA attention XAI by qualified dermatologists is the top priority for future work. A structured validation protocol should be used to present model predictions to board certified dermatologists together with the heatmaps generated by the Grad-CAM, and request them to confirm if the highlighted areas align with the dermoscopic features that they use to make a diagnosis, such as pigment network, vascular networks, border irregularity, and color heterogeneity. If the agreement rate between model attention and expert clinical reasoning was measured, it could be interpreted as a rigorous XAI clinical validation, shifting current visual interpretability to evidentially validated clinical explainability.

Recommendation 2 - Diffusion Model Augmentation: Future work is recommended to use diffusion model based image synthesis in lieu of SSL GAN, such as denoising diffusion probabilistic models (DDPM) or latent diffusion models. While this study shows limited performance with GANs, diffusion models have been able to achieve significantly higher image quality and diversity in medical image synthesis, where the training instability and mode collapse issues faced by GANs were not encountered. Future work on the integrated augmentation strategy could benefit from higher quality synthetic images of minority classes to yield better performance in the model, than seen in the current results.

Recommendation 3 - Multimodal Architecture: For future research, it is suggested to further expand the proposed framework by adding clinical data such as patient age, sex, anatomical location, and lesion duration, which will be combined with dermoscopic image features in a multimodal fusion architecture. The clinical metadata is complementary diagnostic information which cannot be taken from dermoscopic images. In particular, there are strong prior classifiers for some of the lesion types, such as patient age for melanocytic nevus, and BCC is more common in older patients, so incorporating these could help discriminate between some of the visually similar classes.

Recommendation 4 - ISIC 2024 Integration: A large-scale training dataset, the ISIC 2024 (401,058 images), is followed by fine-tuning the 8-class task with the HAM10000 and ISIC 2019 datasets, which is a promising strategy to enhance the model performance and its generalization. This two-stage training strategy would exploit the scale of ISIC 2024 for learning general dermoscopic features of the image and then fine-tune it for the task of adaptation to 8-class classification. This is an approach that should be explored in the future with adequate storage space (around 200GB).

Recommendation 5 - Model Fairness Analysis: More studies are needed on performance of the model in various skin colour groups to understand and mitigate demographic bias. The existing training data sets are biased towards lighter skin tones, which are well represented in European clinical environments, and could lead to lower performance for darker skin tones, which are more prevalent in South Asian populations, such as in Sri Lanka. Incorporating dermoscopic images from various skin tone populations and evaluating the performance of the system by subgroups of skin tone would help to overcome such bias and ensure equitable system deployment.

Recommendation 6 - Mobile and Edge Deployment: For future research, the distribution of the 89.3 million parameter ConvNeXtV2-FSA model to mobile-friendly sizes for deploying the model on smartphones would be beneficial, which can be achieved by using model compression methods like knowledge distillation, quantization, and structured pruning. A mobile-deployable version of the framework would have a significant impact in the clinical world given the rationale of skin cancer screening in resource-limited areas being democratized. The first compression step is to distillate knowledge from a full ConvNextV2-FSA teacher model to a smaller student model, such as EfficientNet-B0 or MobileNet-V3.

Recommendation 7 - Specialist Classifier Improvement: The binary accuracy of the AKIEC versus SCC specialist classifier was 75.09% with significant scope for improvement. Future directions should include: Using more training data for these classes from clinical databases with histopathology confirmation; explicitly using contrastive learning approaches to increase the distance between the representations of AKIEC and SCC; and using dermoscopic metadata, such as the surface of the image (pattern descriptors obtained through image processing) as additional features for the binary decision.

Recommendation 8 - Longitudinal Lesion Monitoring: Further studies are needed to increase the scope of the proposed framework from single-image classification to longitudinal lesion monitoring, to track changes in dermoscopic images of the same lesion over multiple time points to identify early malignant transformation. Also, recurrent or temporal attention mechanisms could be used to model lesion evolution patterns, which has the potential to improve melanoma detection at the earliest stages of melanoma transformation from a benign nevus, the most impactful clinical use case of AI-assisted dermoscopy.

5.5 Final Conclusion

This work has successfully proposed and validated a novel integrated hybrid deep learning framework for multi-class skin cancer early detection that overcomes the three major challenges of current skin cancer early detection system: class imbalance, lack of interpretability, and poor inter-dataset generalization in a single pipeline.

The proposed framework is original in five aspects. Second, the integration of ConvNeXtV2 backbone with Feature Selection Attention is the first reported example of the combination of these two components for the dermoscopic image analysis field, and a hybrid CNN-Attention network that achieves better feature discrimination, but with lower computational complexity than Vision Transformers in clinical applications, has been created. Secondly, the three-component class imbalance strategy used in this work, SSL GAN augmentation, K-means Stratified Sampling and Weighted Random Sampling is more complete than individual techniques in similar literature, addressing the class imbalance issue which is 58:1 in this case. Thirdly, the hierarchical specialist classifier for the discrimination of the AKIEC and SCC task is a novel architectural contribution to the handling of visually similar class pair in multi-class medical image classification. Fourth, unlike existing literature that considers only single method XAI implementations, the multi-faceted XAI framework (Grad-CAM, FSA attention visualization, and t-SNE feature space analysis) offers more comprehensive clinical interpretability. Fifth, the strict cross-dataset validation process, with a maximum performance drop of only 1.79 percentage points on independent test datasets, sets a

new benchmark for evaluation of generalization compared to single-dataset testing found in similar published studies.

The achieved overall accuracy of 91.50% on the combined HAM10000 and ISIC 2019 test set with cross-dataset validation outperforms the published benchmark of Varghese et al. [5] (90.61%) on a more challenging 8-class problem. The hold-out accuracy for the HAM10000 benchmark set is 95.48%, which is a very good generalization to this widely used benchmark set. Inference time of less than 2 seconds on standard GPU hardware makes for practical deployment in clinical settings, and the multi-perspective XAI framework ensures interpretability for the trust and regulatory compliance of clinicians.

The limitations identified were particularly the AKIEC-SCC classification challenge and lack of formal clinical validation of XAI, which are genuine limitations of dermoscopic image-based AI diagnosis, and not flaws in the proposed approach. This study's contributions provide a solid foundation for future research, given these limitations.

In conclusion, the research shows that by integrating state-of-the-art CNN architecture, attention-based feature selection, extensive class imbalance mitigation, and multi-perspective explainability, a clinically relevant, computationally efficient, and truly generalizable skin cancer detection system can be achieved. The open source distribution of the model weights and implementation code will allow the research community to draw on these contributions and help move the collective effort toward life-saving, timely, accurate and interpretable diagnosis for early detection of skin cancer forward.

REFERENCES

[1] American Cancer Society, "Cancer Facts & Figures 2024," American Cancer Society, Atlanta, GA, 2024.

[2] A. C. Society, "Survival Rates for Melanoma Skin Cancer," 2024. [Online]. Available: https://www.cancer.org/cancer/melanoma-skin-cancer/detection-diagnosis-staging/survival-rates-for-melanoma-skin-cancer-by-stage.html

[3] K. Safdar, S. Akbar, U. Shahzore, S. Iqbal, N. Ayesha, and S. Gull, "Explainable AI for melanoma diagnosis through dermoscopic images: Recent findings and future directions," in Explainable AI in Healthcare, Elsevier, 2025, pp. 245–268.

[4] A. Esteva et al., "Dermatologist-level classification of skin cancer with deep neural networks," Nature, vol. 542, no. 7639, pp. 115–118, Feb. 2017.

[5] A. Varghese et al., "Robust Skin Cancer Detection through CNN-Transformer-GRU Fusion and Generative Adversarial Network Based Data Augmentation," Computer Modeling in Engineering & Sciences, vol. 142, no. 3, pp. 2891–2915, 2025.

[6] M. Bhargavi and S. Balakrishna, "Transfer learning based hybrid feature learning framework for enhanced skin cancer diagnosis using deep feature integration," Biomedical Signal Processing and Control, vol. 98, p. 106955, Jan. 2025.

[7] M. Alzamel, C. Iliopoulos, and Z. Lim, "Deep learning approaches and data augmentation for melanoma detection," Neural Computing and Applications, vol. 36, pp. 12469–12485, 2024.

[8] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," Scientific Data, vol. 5, p. 180161, Aug. 2018.

[9] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," International Journal of Computer Vision, vol. 128, no. 2, pp. 336–359, Feb. 2020.

[10] A. G. C. Pacheco et al., "PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones," Data in Brief, vol. 32, p. 106221, Oct. 2020.

[11] J. M. Chaudhary, G. N. Nadeem, and M. S. Qureshi, "Telemedicine and artificial intelligence in dermatology during COVID-19 pandemic," Journal of Medical Systems, vol. 45, no. 8, p. 78, Aug. 2021.

[12] R. L. Siegel, K. D. Miller, N. S. Wagle, and A. Jemal, "Cancer statistics, 2023," CA: A Cancer Journal for Clinicians, vol. 73, no. 1, pp. 17–48, Jan. 2023.

[13] W. Stolz et al., "ABCD rule of dermatoscopy: A new practical method for early recognition of malignant melanoma," European Journal of Dermatology, vol. 4, pp. 521–527, 1994.

[14] N. R. Telfer, G. B. Colver, and C. A. Morton, "Guidelines for the management of basal cell carcinoma," British Journal of Dermatology, vol. 159, no. 1, pp. 35–48, 2008.

[15] W. M. S. Wickramasinghe, L. Perera, and K. De Silva, "Application of artificial intelligence to education in Sri Lanka," Sri Lanka Guardian, Nov. 2024.

[16] H. A. Haenssle et al., "Man against machine: diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists," Annals of Oncology, vol. 29, no. 8, pp. 1836–1842, Aug. 2018.

[17] N. C. F. Codella et al., "Skin Lesion Analysis Toward Melanoma Detection 2018: A Challenge Hosted by the International Skin Imaging Collaboration," arXiv preprint arXiv:1902.03368, 2019.

[18] N. C. F. Codella et al., "Skin Lesion Analysis Toward Melanoma Detection: ISIC 2019 Challenge," arXiv, 2019.

[19] D. Meedeniya, S. De Silva, L. Gamage, and U. Isuranga, "Skin cancer identification utilizing deep learning: A survey," IET Image Processing, vol. 18, pp. 3225–3245, 2024.

[20] S. Adamu et al., "The future of skin cancer diagnosis: a comprehensive systematic literature review of machine learning and deep learning models," Cogent Engineering, vol. 11, no. 1, p. 2395425, 2024.

[21] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE CVPR, 2016, pp. 770–778.

[22] A. Rehman Khan et al., "Early-Stage Melanoma Cancer Diagnosis Framework for Imbalanced Data From Dermoscopic Images," Microscopy Research and Technique, vol. 87, pp. 2736–2750, 2024.

[23] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in Proc. IEEE CVPR, 2017, pp. 4700–4708.

[24] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. ICML, 2019, pp. 6105–6114.

[25] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in Proc. ICLR, 2021.

[26] A. Radford et al., "Learning transferable visual models from natural language supervision," in Proc. ICML, 2021, pp. 8748–8763.

[27] Z. Liu et al., "A ConvNet for the 2020s," in Proc. IEEE CVPR, 2022, pp. 11976–11986.

[28] S. Woo et al., "ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders," in Proc. IEEE/CVF CVPR, 2023, pp. 16133–16142.

[29] P. Silva et al., "ConvNeXt-based architectures for skin cancer classification: A comparative study," Medical Image Analysis, vol. 78, p. 102401, May 2022.

[30] Y. Gautam et al., "FusionEXNet: an interpretable fused deep learning model for skin cancer detection," Cogent Engineering, vol. 11, no. 1, p. 2395425, 2024.

[31] M. D. Nayeem et al., "Multi-Scale Attention Fusion With Depthwise Separable Convolutions for Efficient Skin Cancer Detection," Computational Intelligence, vol. 40, no. 1, p. e14870, 2024.

[32] S. S. Abraham and J. D. Peter, "Retrospective Analysis of an Interpretable Skin Cancer Classification using Deep Learning Models," in Proc. ICRAI, 2024, pp. 1–6.

[33] A. M. Rifa'i et al., "Leveraging Data Augmentation and Dropout Layer in MobileNetV3 for Accurate Skin Cancer Detection ISIC Dataset," in Proc. ICITISEE, 2024, pp. 462–467.

[34] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 42, no. 2, pp. 318–327, Feb. 2020.

[35] L. Chen, Y. Wang, and H. Zhang, "Cluster-based sampling strategies for imbalanced skin cancer datasets," Journal of Biomedical Informatics, vol. 145, p. 104457, Sep. 2024.

[36] I. Goodfellow et al., "Generative adversarial nets," in Proc. NeurIPS, 2014, pp. 2672–2680.

[37] A. Bissoto, M. Fornaciali, E. Valle, and S. Avila, "Skin lesion synthesis with generative adversarial networks," in OR 2.0 Context-Aware Operating Theaters, Springer, 2018, pp. 294–302.

[38] Y. Liu, J. Chen, and Z. Wang, "Semi-supervised learning with generative adversarial networks for skin lesion classification," Medical Physics, vol. 47, no. 11, pp. 5413–5424, Nov. 2020.

[39] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," in Proc. ICLR, 2018.

[40] M. Siva Sibi and J. Anitha, "Skin Cancer Detection in 3-Dimensional Total Body Photography using Deep Learning," IEEE Access, vol. 10, pp. 112468–112483, 2022.

[41] N. H. Sree et al., "Explainable AI Insights into Skin Cancer Detection: A Comparative Study of CNN, DenseNet, and ResNet," in Proc. I2CT, 2024, pp. 1–6.

[42] A. Rajkomar, J. Dean, and I. Kohane, "Machine learning in medicine," New England Journal of Medicine, vol. 380, no. 14, pp. 1347–1358, Apr. 2019.

[43] G. Munjal et al., "SkinSage XAI: An explainable deep learning solution for skin lesion diagnosis," Health Care Science, vol. 3, pp. 237–248, 2024.

[44] S. S. Abraham and J. D. Peter, "Retrospective Analysis of an Interpretable Skin Cancer Classification using Deep Learning Models," in Proc. ICRAI, 2024, pp. 1–6.

[45] A. Alrabai, A. Echtioui, and F. Kallel, "Explainable deep learning approaches for skin cancer diagnosis," Neural Computing and Applications, vol. 36, pp. 20413–20432, 2024.

[46] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," Journal of Machine Learning Research, vol. 9, pp. 2579–2605, Nov. 2008.

[47] J. Deng et al., "ImageNet: A large-scale hierarchical image database," in Proc. IEEE CVPR, 2009, pp. 248–255.

[48] Y. Ganin et al., "Domain-adversarial training of neural networks," Journal of Machine Learning Research, vol. 17, no. 59, pp. 1–35, 2016.

[49] T. Lee, V. Ng, R. Gallagher, A. Coldman, and D. McLean, "Dullrazor: A software approach to hair removal from images," *Computers in Biology and Medicine*, vol. 27, no. 6, pp. 533–543, 1997.

[50] A. Telea, "An image inpainting technique based on the fast marching method," *Journal of Graphics Tools*, vol. 9, no. 1, pp. 23–34, 2004.

Appendix A: ConvNeXtV2 + FSA Model Code

```python
import torch
import torch.nn as nn
import timm

class FeatureSelectionAttention(nn.Module):
    """
    Feature Selection Attention Module
    Sequential: Channel Attention → Spatial Attention
    Complexity: O(n) vs O(n²) for full self-attention
    """
    def __init__(self, in_channels, reduction=16):
        super().__init__()
        # Channel attention components
        self.channel_avg = nn.AdaptiveAvgPool2d(1)
        self.channel_max = nn.AdaptiveMaxPool2d(1)
        self.channel_fc  = nn.Sequential(
            nn.Linear(
                in_channels,
                in_channels // reduction,
                bias=False
            ),
            nn.ReLU(),
            nn.Linear(
                in_channels // reduction,
                in_channels,
                bias=False
            )
        )
        # Spatial attention components
        self.spatial_conv = nn.Conv2d(
            2, 1, kernel_size=7, padding=3, bias=False
        )
        self.sigmoid = nn.Sigmoid()

    def forward(self, x):
        B, C, H, W = x.shape

        # Channel Attention: Ac
        avg = self.channel_fc(
            self.channel_avg(x).view(B, C)
        )
        mx  = self.channel_fc(
            self.channel_max(x).view(B, C)
        )
        ca  = self.sigmoid(avg + mx).view(B, C, 1, 1)
        x   = x * ca  # X' = X ⊗ Ac

        # Spatial Attention: As
        avg_sp = torch.mean(x, dim=1, keepdim=True)
        max_sp, _ = torch.max(x, dim=1, keepdim=True)
        sa  = self.sigmoid(
            self.spatial_conv(
                torch.cat([avg_sp, max_sp], dim=1)
            )
        )
        x = x * sa  # X'' = X' ⊗ As
        return x
```

```python
class ConvNeXtV2WithFSA(nn.Module):
    """
    Proposed Hybrid Architecture:
    ConvNeXtV2-Base + FSA + Classification Head
    Total parameters: ~89.3M
    """
    def __init__(self, num_classes=8, dropout_rate=0.4):
        super().__init__()
        # ConvNeXtV2 backbone
        self.backbone = timm.create_model(
            'convnextv2_base',
            pretrained=True,
            num_classes=0,
            global_pool=''
        )
        self.feature_dim = self.backbone.num_features
        # 1024 for convnextv2_base

        # FSA module
        self.fsa = FeatureSelectionAttention(
            self.feature_dim, reduction=16
        )

        # Global average pooling
        self.gap = nn.AdaptiveAvgPool2d(1)

        # Classification head
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(self.feature_dim, 512),
            nn.BatchNorm1d(512),
            nn.ReLU(),
            nn.Dropout(dropout_rate),
            nn.Linear(512, 256),
            nn.BatchNorm1d(256),
            nn.ReLU(),
            nn.Dropout(dropout_rate / 2),
            nn.Linear(256, num_classes)
        )

    def forward(self, x):
        features = self.backbone(x)   # [B,1024,7,7]
        features = self.fsa(features) # Attention applied
        features = self.gap(features) # [B,1024,1,1]
        return self.classifier(features)  # [B, 8]
```

Appendix B: Complete Ablation Study Results

```python
# ── EXPERIMENT 1: Baseline (no FSA, no GAN, no clustering) ──
print("\n" + "="*55)
print("EXPERIMENT 1: Baseline ConvNeXtV2 (no FSA)")
print("Training data: original combined_train (no GAN)")
print("="*55)

# Use original training data WITHOUT GAN synthetic images
# Filter to remove any 'augmented' source images if present
if 'source' in train_original.columns:
    train_exp1 = train_original[
        train_original['source'] == 'real'
    ].copy()
else:
    train_exp1 = train_original.copy()

print(f"Training samples: {len(train_exp1)}")
print(train_exp1['dx'].value_counts())

tr1, vl1, ts1 = make_loaders(train_exp1)
model1 = ConvNeXtV2Baseline(NUM_CLASSES).to(device)
ablation_results['exp1'] = train_ablation_model(
    model1, tr1, vl1, ts1,
    "Exp1: ConvNeXtV2 Baseline (no FSA, no GAN)",
    epochs=25, lr=3e-5
)

# Save checkpoint
torch.save({
    'model_state_dict': model1.state_dict(),
    'test_acc': ablation_results['exp1']['test_acc']
}, 'E:/SkinCancerResearch/models/ablation_exp1.pth')
print(f"\nExp1 saved! Test acc: {ablation_results['exp1']['test_acc']:.2f}%")
```

```python
# ── EXPERIMENT 2: Add FSA module ──
print("\n" + "="*55)
print("EXPERIMENT 2: ConvNeXtV2 + FSA (no GAN)")
print("Training data: original (no GAN synthetic)")
print("="*55)

# ── SAFETY CHECK: recreate train_exp1 if missing ──
if 'train_exp1' not in globals():
    print("train_exp1 not found. Reconstructing from train_original...")

    if 'source' in train_original.columns:
        train_exp1 = train_original[train_original['source'] == 'real'].copy()
    else:
        train_exp1 = train_original.copy()

# ── Loaders ──
tr2, vl2, ts2 = make_loaders(train_exp1)

# ── Model ──
model2 = ConvNeXtV2WithFSA(NUM_CLASSES).to(device)

# ── Training ──
ablation_results['exp2'] = train_ablation_model(
    model2, tr2, vl2, ts2,
    "Exp2: ConvNeXtV2 + FSA (no GAN)",
    epochs=25, lr=3e-5
)

# ── Save model ──
torch.save({
    'model_state_dict': model2.state_dict(),
    'test_acc': ablation_results['exp2']['test_acc']
}, 'E:/SkinCancerResearch/models/ablation_exp2.pth')

print(f"\nExp2 saved! Test acc: {ablation_results['exp2']['test_acc']:.2f}%")
print(
    f"FSA contribution: +"
    f"{ablation_results['exp2']['test_acc'] - ablation_results['exp1']['test_acc']:.2f}%"
)
```

```
# ── EXPERIMENT 3: Add GAN augmentation ──
print("\n" + "="*55)
print("EXPERIMENT 3: ConvNeXtV2 + FSA + GAN")
print("Training data: with GAN synthetic images")
print("="*55)

# Use GAN augmented training data
train_exp3 = pd.read_csv(
    'E:/SkinCancerResearch/data/final_train_with_gan.csv'
)

# Reduce nv to avoid dominance
nv_df      = train_exp3[train_exp3['dx']=='nv'].sample(
    n=3500, random_state=42
)
other_df   = train_exp3[train_exp3['dx']!='nv']
train_exp3 = pd.concat(
    [nv_df, other_df], ignore_index=True
).sample(frac=1, random_state=42)

print(f"Training samples: {len(train_exp3)}")
print(train_exp3['dx'].value_counts())

tr3, vl3, ts3 = make_loaders(train_exp3)
model3 = ConvNeXtV2WithFSA(NUM_CLASSES).to(device)
ablation_results['exp3'] = train_ablation_model(
    model3, tr3, vl3, ts3,
    "Exp3: ConvNeXtV2 + FSA + GAN",
    epochs=25, lr=3e-5
)

torch.save({
    'model_state_dict': model3.state_dict(),
    'test_acc': ablation_results['exp3']['test_acc']
}, 'E:/SkinCancerResearch/models/ablation_exp3.pth')
print(f"\nExp3 saved! Test acc: {ablation_results['exp3']['test_acc']:.2f}%")
print(f"GAN contribution: +{ablation_results['exp3']['test_acc']-ablation_results['exp2']['test_acc']:.2f}%")
```

```
# ── EXPERIMENT 4: Add clustering-based sampling ──
print("\n" + "="*55)
print("EXPERIMENT 4: ConvNeXtV2 + FSA + GAN + Clustering")
print("Training data: clustered balanced dataset")
print("="*55)

train_exp4 = pd.read_csv(
    'E:/SkinCancerResearch/data/final_train_reduced.csv'
)
print(f"Training samples: {len(train_exp4)}")
print(train_exp4['dx'].value_counts())

tr4, vl4, ts4 = make_loaders(train_exp4)
model4 = ConvNeXtV2WithFSA(NUM_CLASSES).to(device)
ablation_results['exp4'] = train_ablation_model(
    model4, tr4, vl4, ts4,
    "Exp4: ConvNeXtV2 + FSA + GAN + Clustering",
    epochs=25, lr=3e-5
)

torch.save({
    'model_state_dict': model4.state_dict(),
    'test_acc': ablation_results['exp4']['test_acc']
}, 'E:/SkinCancerResearch/models/ablation_exp4.pth')
print(f"\nExp4 saved! Test acc: {ablation_results['exp4']['test_acc']:.2f}%")
print(f"Clustering contribution: +{ablation_results['exp4']['test_acc']-ablation_results['exp3']['test_acc']:.2f}%")
```

```python
# ── EXPERIMENT 5: Full proposed model ──
print("\n" + "="*55)
print("EXPERIMENT 5: Full Model + Specialist Classifier")
print("Loading best trained model from previous work")
print("="*55)

# Load our best trained model
best_model = ConvNeXtV2WithFSA(NUM_CLASSES).to(device)
ckpt = torch.load(
    'E:/SkinCancerResearch/models/convnextv2_gan_finetuned.pth',
    map_location=device, weights_only=False
)
best_model.load_state_dict(ckpt['model_state_dict'])

# Load specialist
class SpecialistClassifier(nn.Module):
    def __init__(self, dropout=0.4):
        super().__init__()
        self.backbone = timm.create_model(
            'efficientnet_b3', pretrained=False,
            num_classes=0, global_pool='avg'
        )
        feat_dim = self.backbone.num_features
        self.classifier = nn.Sequential(
            nn.Linear(feat_dim, 256),
            nn.BatchNorm1d(256),
            nn.ReLU(),
            nn.Dropout(dropout),
            nn.Linear(256, 64),
            nn.ReLU(),
            nn.Dropout(dropout/2),
            nn.Linear(64, 2)
        )

    def forward(self, x):
        return self.classifier(self.backbone(x))

specialist = SpecialistClassifier().to(device)
spec_ckpt  = torch.load(
    'E:/SkinCancerResearch/models/specialist_akiec_scc.pth',
    map_location=device, weights_only=False
)
specialist.load_state_dict(spec_ckpt['model_state_dict'])
```

```python
# Cascaded evaluation
akiec_idx = label_to_idx['akiec']
scc_idx   = label_to_idx['scc']

_, _, ts5 = make_loaders(train_exp4)

best_model.eval()
specialist.eval()
all_preds, all_labels = [], []

with torch.no_grad():
    for imgs, lbls in tqdm(ts5, desc="Full model eval"):
        imgs = imgs.to(device)
        out  = best_model(imgs)
        pred = out.argmax(1)

        # Specialist refinement
        amb  = (pred == akiec_idx) | (pred == scc_idx)
        if amb.sum() > 0:
            sp_out  = specialist(imgs[amb])
            sp_pred = sp_out.argmax(1)
            refined = torch.where(
                sp_pred == 0,
                torch.tensor(akiec_idx).to(device),
                torch.tensor(scc_idx).to(device)
            )
            pred[amb] = refined

        all_preds.extend(pred.cpu().numpy())
        all_labels.extend(lbls.numpy())

all_preds  = np.array(all_preds)
all_labels = np.array(all_labels)
test_acc5  = (all_preds == all_labels).mean() * 100

class_accs5 = {}
for i in range(NUM_CLASSES):
    mask = all_labels == i
    if mask.sum() > 0:
        class_accs5[idx_to_label[i]] = (
            all_preds[mask] == i
        ).mean() * 100

ablation_results['exp5'] = {
    'name'      : 'Exp5: Full Model + Specialist',
    'val_acc'   : ckpt['val_acc'],
    'test_acc'  : test_acc5,
    'preds'     : all_preds,
    'labels'    : all_labels,
    'class_accs': class_accs5
}
print(f"\nExp5 Test Acc: {test_acc5:.2f}%")
print(f"Specialist contribution: +{test_acc5-ablation_results['exp4']['test_acc']:.2f}%")
```

Appendix C: Full Cross-dataset Results

```python
cross_results = {}

# ── Evaluation 1: Combined test set ──
print("\n" + "="*55)
print("EVALUATION 1: Combined Test Set")
print("(HAM10000 + ISIC 2019 combined)")
print("="*55)

combined_test = pd.read_csv(
    'E:/SkinCancerResearch/data/combined_test.csv'
)
cross_results['combined'] = evaluate_dataset(
    model, combined_test,
    "Combined Test (HAM10000 + ISIC 2019)",
    device, label_to_idx, idx_to_label
)

# ── Evaluation 2: HAM10000 only ──
print("\n" + "="*55)
print("EVALUATION 2: HAM10000 Hold-out Test")
print("(Independent dataset — different distribution)")
print("="*55)

cross_results['ham10000'] = evaluate_dataset(
    model, ham_test,
    "HAM10000 Hold-out",
    device, label_to_idx, idx_to_label
)

# ── Evaluation 3: ISIC 2019 only ──
print("\n" + "="*55)
print("EVALUATION 3: ISIC 2019 Hold-out Test")
print("="*55)

cross_results['isic2019'] = evaluate_dataset(
    model, isic_test_only,
    "ISIC 2019 Hold-out",
    device, label_to_idx, idx_to_label
)

# Summary
print("\n" + "="*55)
print("CROSS-DATASET SUMMARY")
print("="*55)
for key, r in cross_results.items():
    print(f"  {r['name']:40s}: {r['overall']:.2f}%")
```

```
=======================================================
EVALUATION 1: Combined Test Set
(HAM10000 + ISIC 2019 combined)
=======================================================
Evaluating Combined Test (HAM10000 + ISIC 2019): 100%|████████████████████████| 166/166 [00:52<00:00,  3.13it/s]

Combined Test (HAM10000 + ISIC 2019) Results:
  Overall Accuracy: 91.34%
  Samples         : 5303

=======================================================
EVALUATION 2: HAM10000 Hold-out Test
(Independent dataset — different distribution)
=======================================================
Evaluating HAM10000 Hold-out: 100%|████████████████████████████████████████| 47/47 [00:10<00:00,  4.54it/s]

HAM10000 Hold-out Results:
  Overall Accuracy: 95.48%
  Samples         : 1503

=======================================================
EVALUATION 3: ISIC 2019 Hold-out Test
=======================================================
Evaluating ISIC 2019 Hold-out: 100%|████████████████████████████████████████| 119/119 [00:24<00:00,  4.81it/s]

ISIC 2019 Hold-out Results:
  Overall Accuracy: 89.71%
  Samples         : 3800

=======================================================
CROSS-DATASET SUMMARY
=======================================================
  Combined Test (HAM10000 + ISIC 2019)    : 91.34%
  HAM10000 Hold-out                       : 95.48%
  ISIC 2019 Hold-out                      : 89.71%
```